

Linear Model Selection and Regularization

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Why consider alternatives to least squares?

- **Prediction Accuracy**: especially when $p > n$, to control the variance.
- **Model Interpretability**: By removing irrelevant features — that is, by setting the corresponding coefficient estimates to zero — we can obtain a model that is more easily interpreted.

Three classes of methods

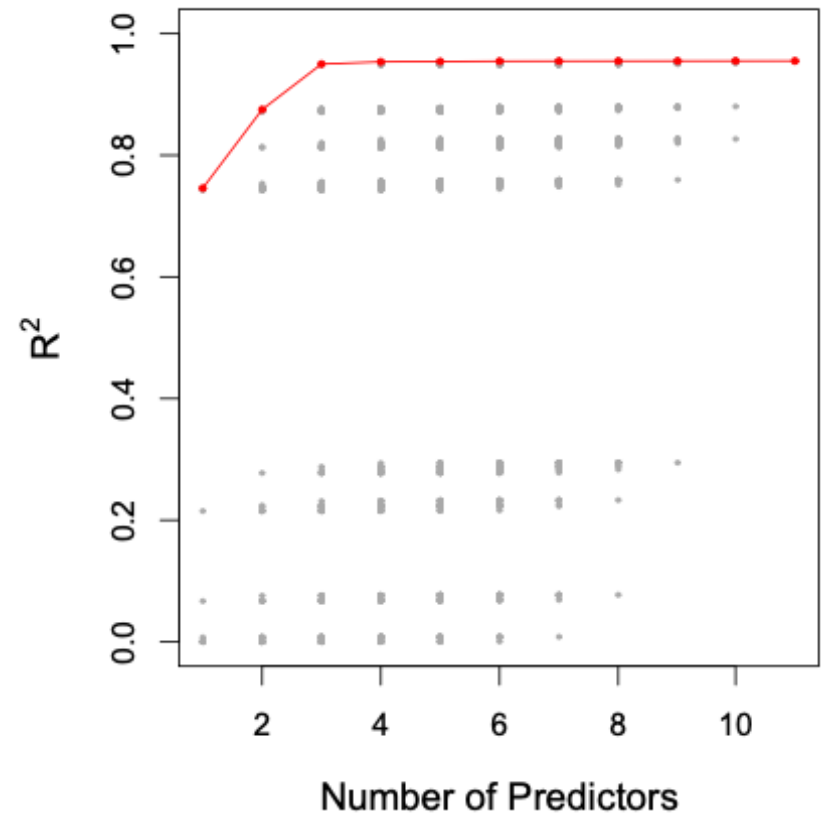
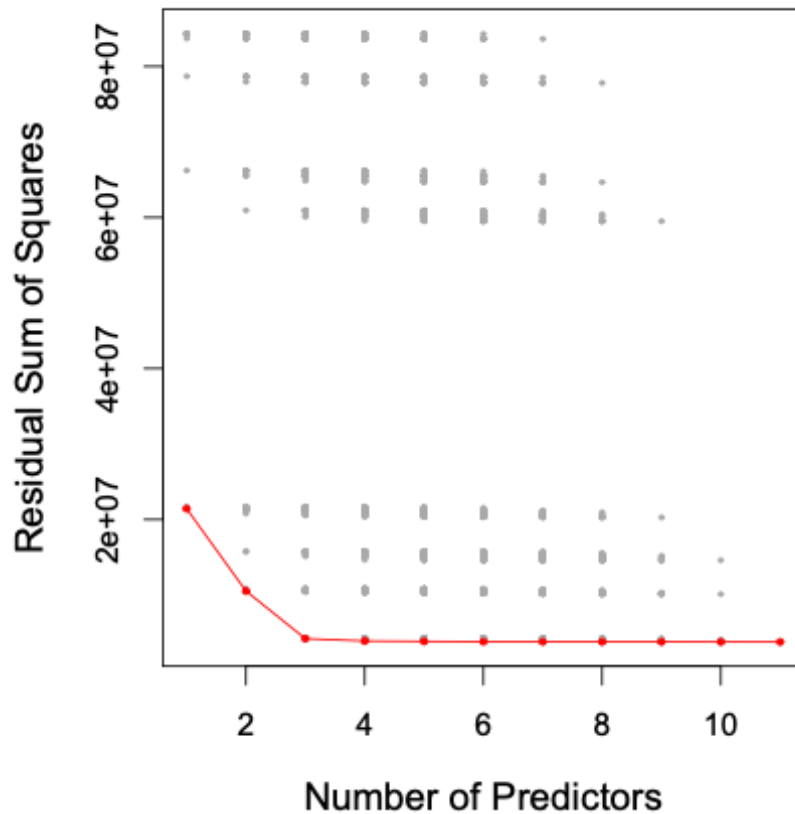
1. **Subset selection:** identify a subset of the p predictors
 - Discrete process
2. **Shrinkage also known as regularization:** fit a model involving all p predictors, but the estimated coefficients are shrunk towards zero
 - Continuous process
3. **Dimension reduction:** We project the p predictors into a M -dimensional subspace, where $M < p$. Then these M projections are used as predictors to fit a linear regression model by least squares.

1.Subset selection

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
 2. For $k = 1, 2, \dots, p$:
 - (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
 - (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2 .
 3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2 . Or use the cross-validation method.
-

Possible model containing a subset of the ten predictors in data set



Stepwise Selection

- For computational reasons, best subset selection cannot be applied with very large p .
- Best subset selection may also suffer from statistical problems when p is large: larger the search space, the higher the chance of finding models that look good on the training data.
- Thus an enormous search space can lead to **overfitting** and high variance of the coefficient estimates.

Forward Stepwise Selection

- Forward stepwise selection begins with a model containing no predictors, and then adds predictors to the model, one-at-a-time, until all of the predictors are in the model.
- In particular, at each step the variable that gives the greatest **additional** improvement to the fit is added to the model.

Subset selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
 2. For $k = 0, \dots, p - 1$:
 - (a) Consider all $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
 - (b) Choose the *best* among these $p - k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2 .
 3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2 . Or use the cross-validation method.
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More on Forward Stepwise Selection

- Computational advantage over best subset selection is clear.
- It is not guaranteed to find the best possible model out of all 2^p models containing subsets of the p predictors. **Why not?**
Give an example.

More on Forward Stepwise Selection

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income student, limit	rating, income, student, limit

- The first four selected models for best subset selection and forward stepwise selection on the Credit data set. The first three models are identical but the fourth models differ.

Subset selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
 2. For $k = p, p - 1, \dots, 1$:
 - (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k - 1$ predictors.
 - (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2 .
 3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2 . Or use the cross-validation method.
-

Choosing the Optimal Model

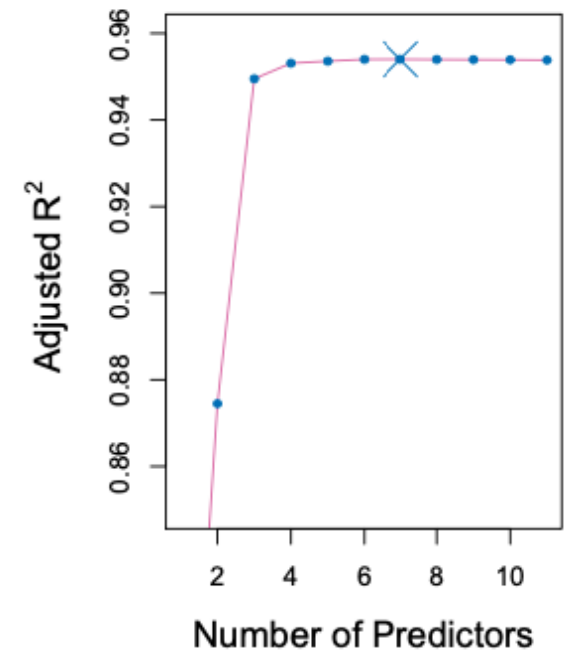
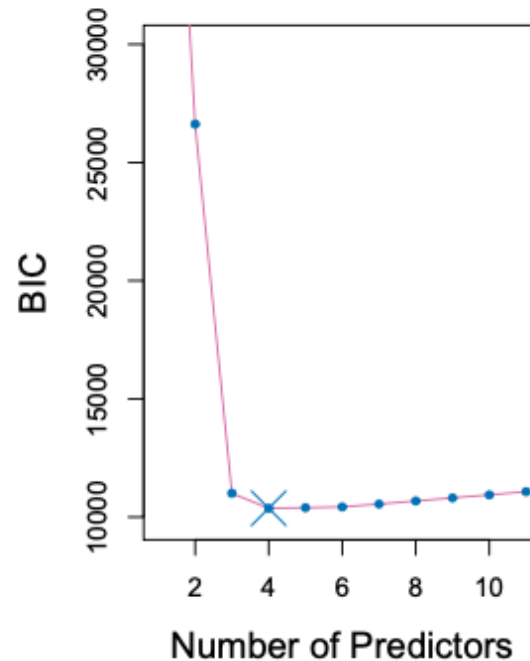
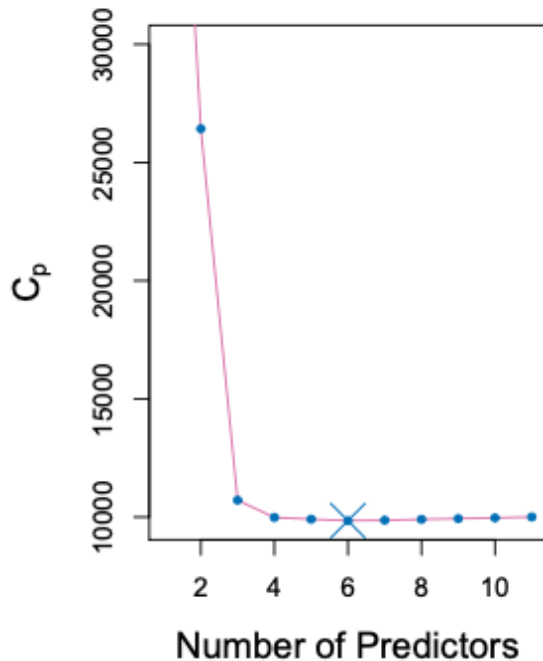
- The model containing all of the predictors will always have the **smallest RSS** and the **largest R^2** , since these quantities are related to the **training error**.
- We wish to choose a model with low test error, not a model with low training error.
- Therefore, **RSS** and **R^2** are not suitable for selecting the best model among a collection of models with different numbers of predictors.

Estimating test error: two approaches

- We can indirectly estimate test error by making an **adjustment** to the training error to account for the bias due to overfitting.
- We can **directly** estimate the test error, using either a validation set approach or a cross-validation approach, as discussed in previous lectures.

C_p , AIC, BIC, and Adjusted R^2

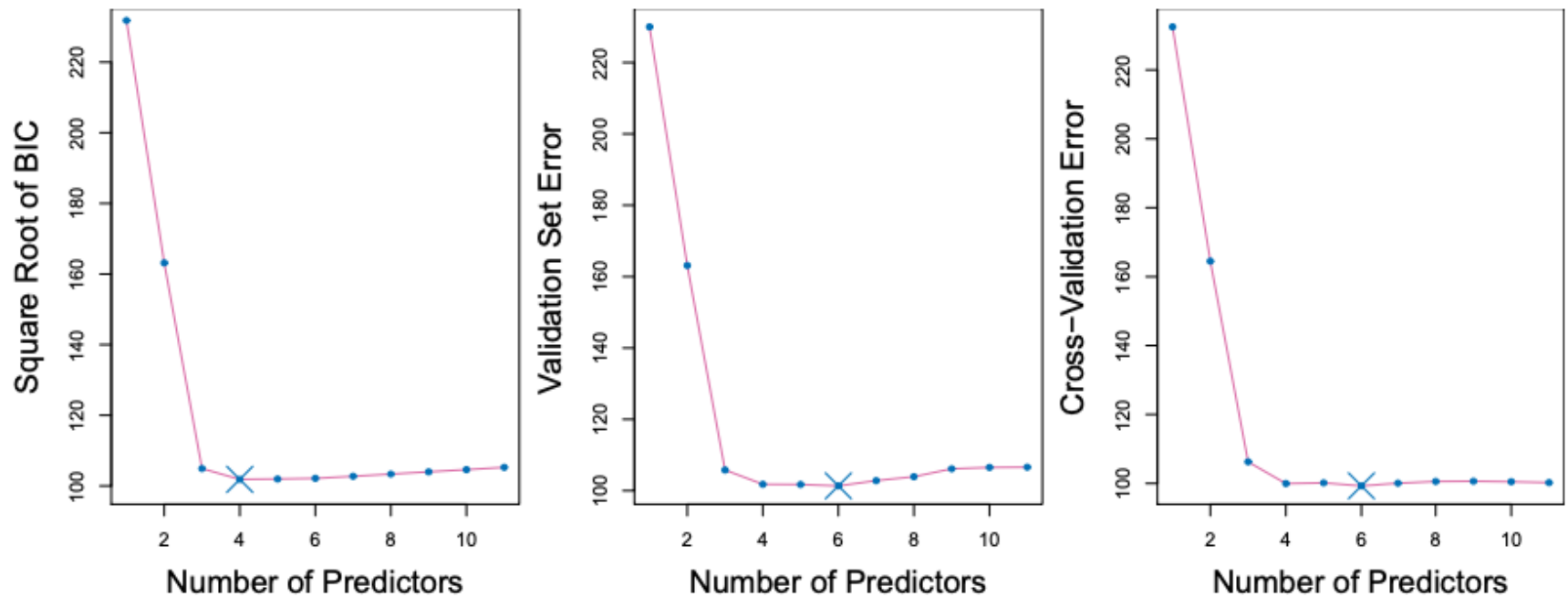
$$\text{BIC} = \frac{1}{n} (\text{RSS} + \log(n)d\hat{\sigma}^2)$$



$$C_p = \frac{1}{n} (\text{RSS} + 2d\hat{\sigma}^2)$$

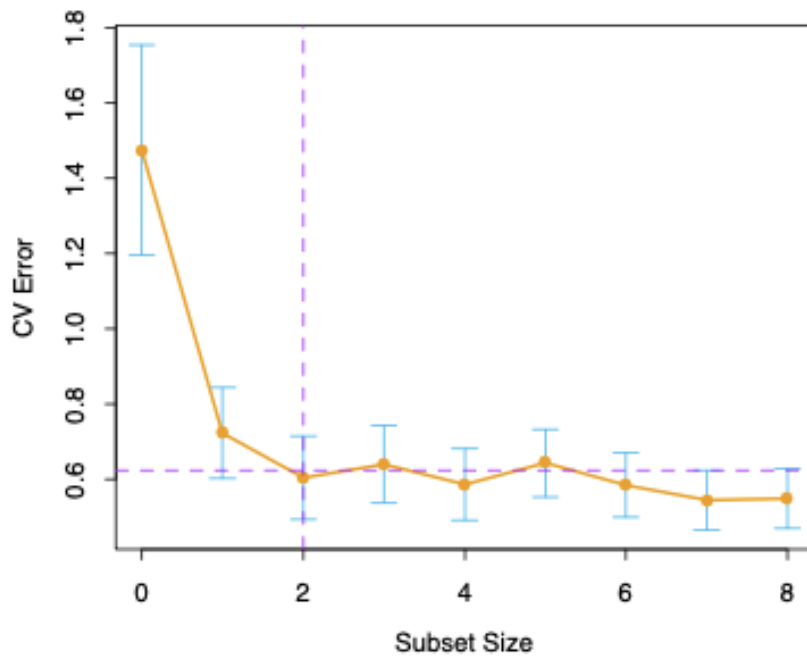
$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS}/(n - d - 1)}{\text{TSS}/(n - 1)}$$

Validation and Cross-Validation

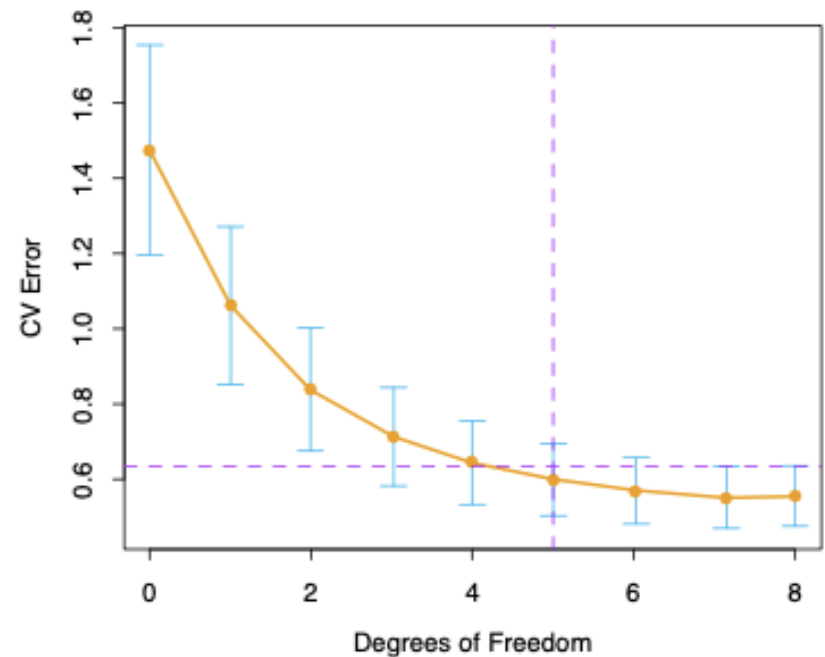


One-standard-error rule

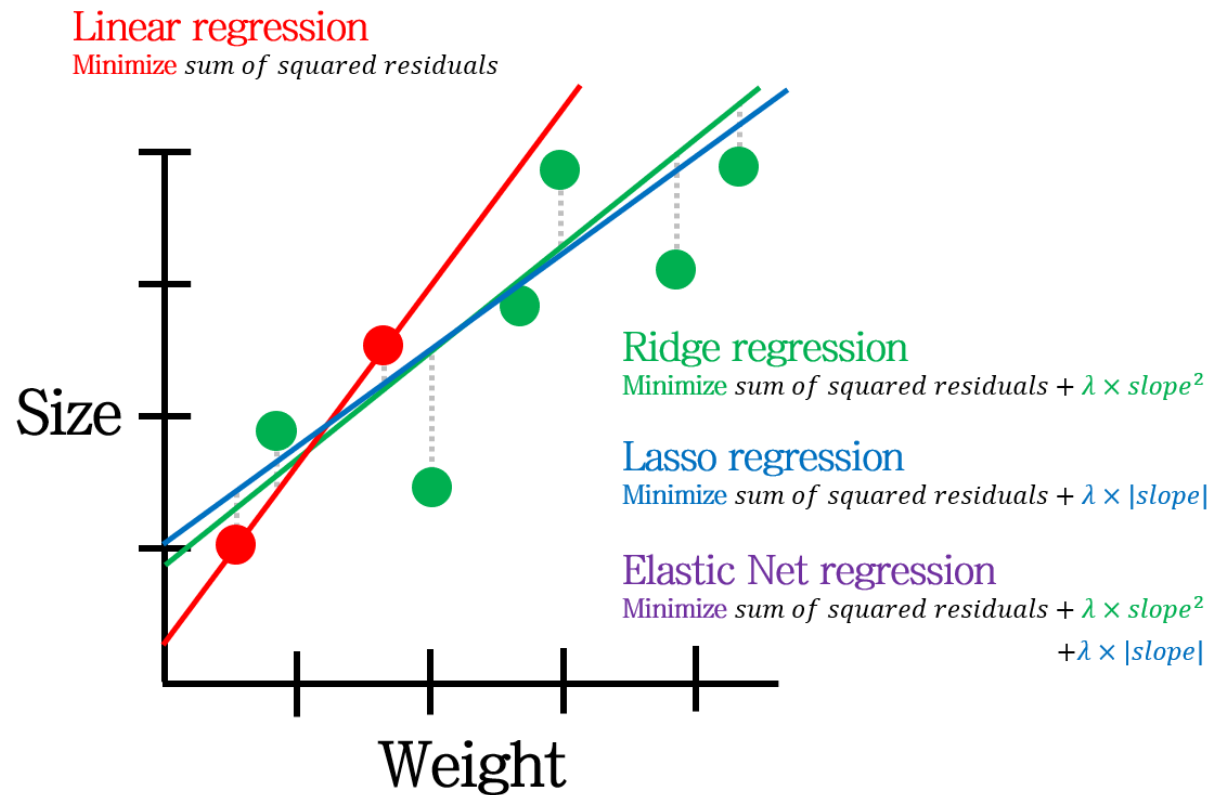
All Subsets



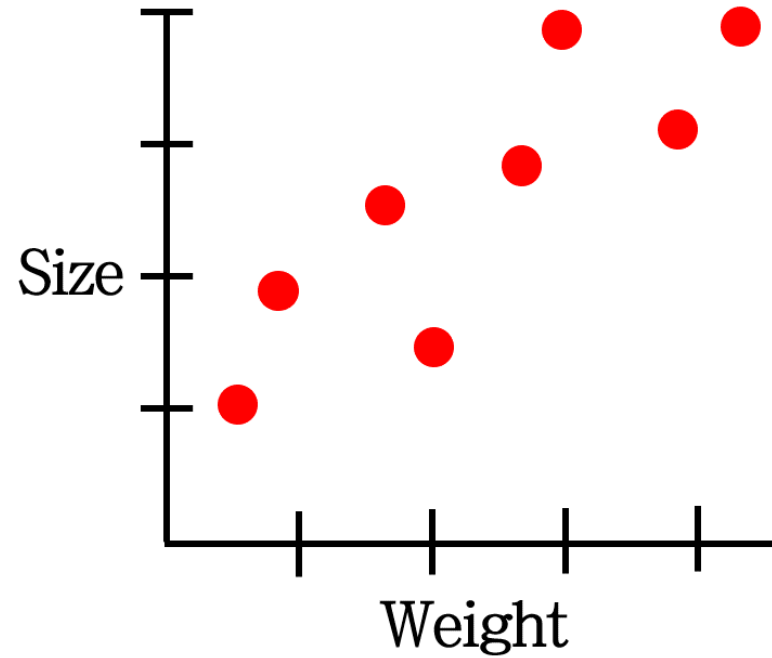
Ridge Regression



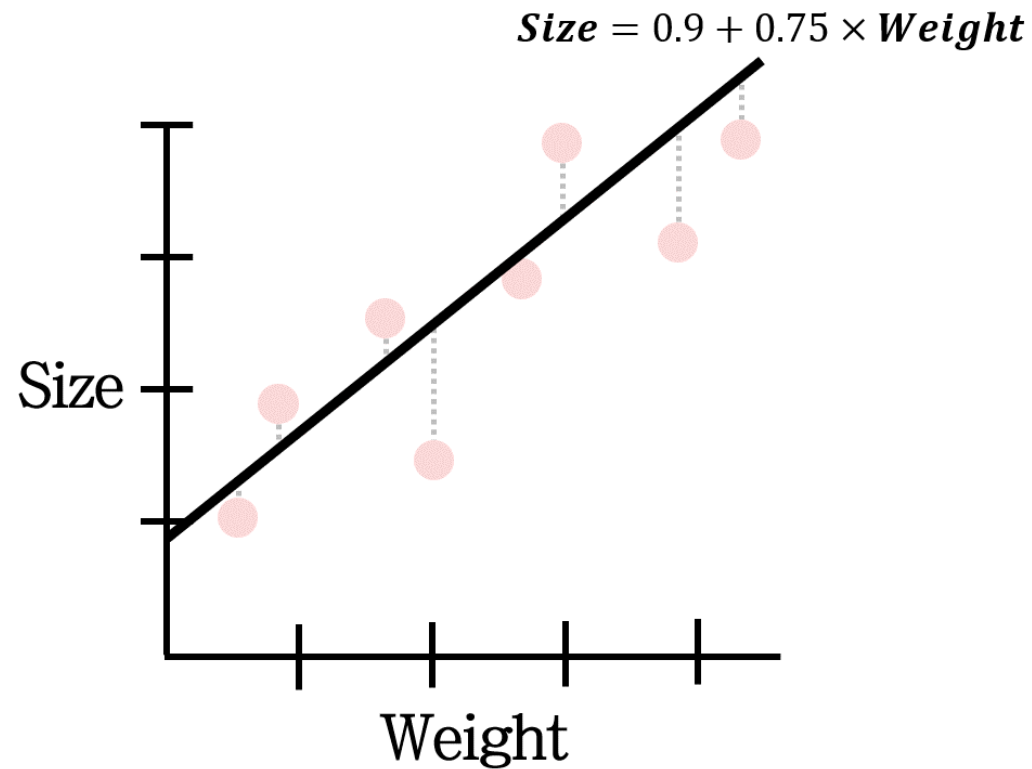
2. Shrinkage also known as regularization



Linear Regression

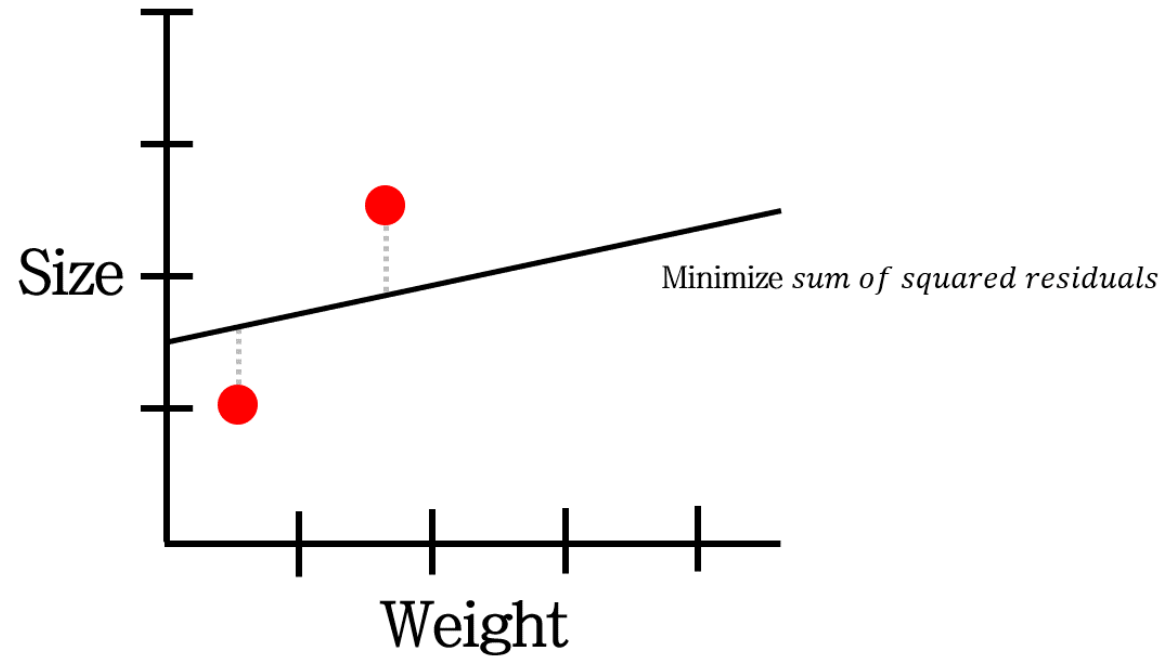


Linear Regression

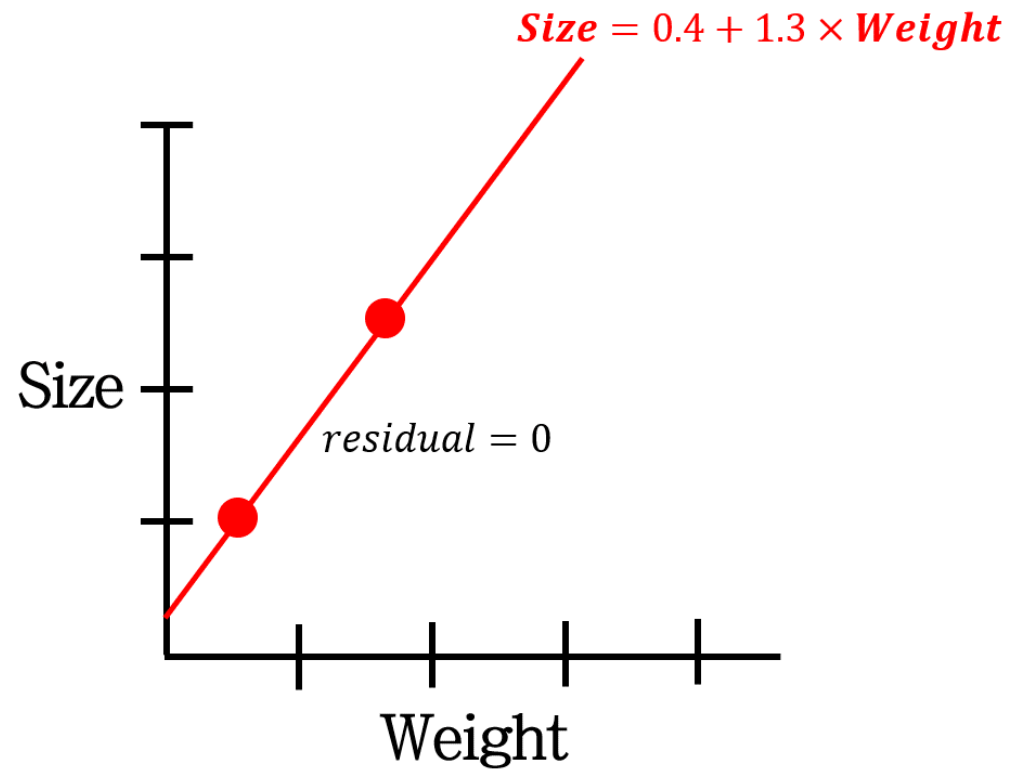


Linear Regression

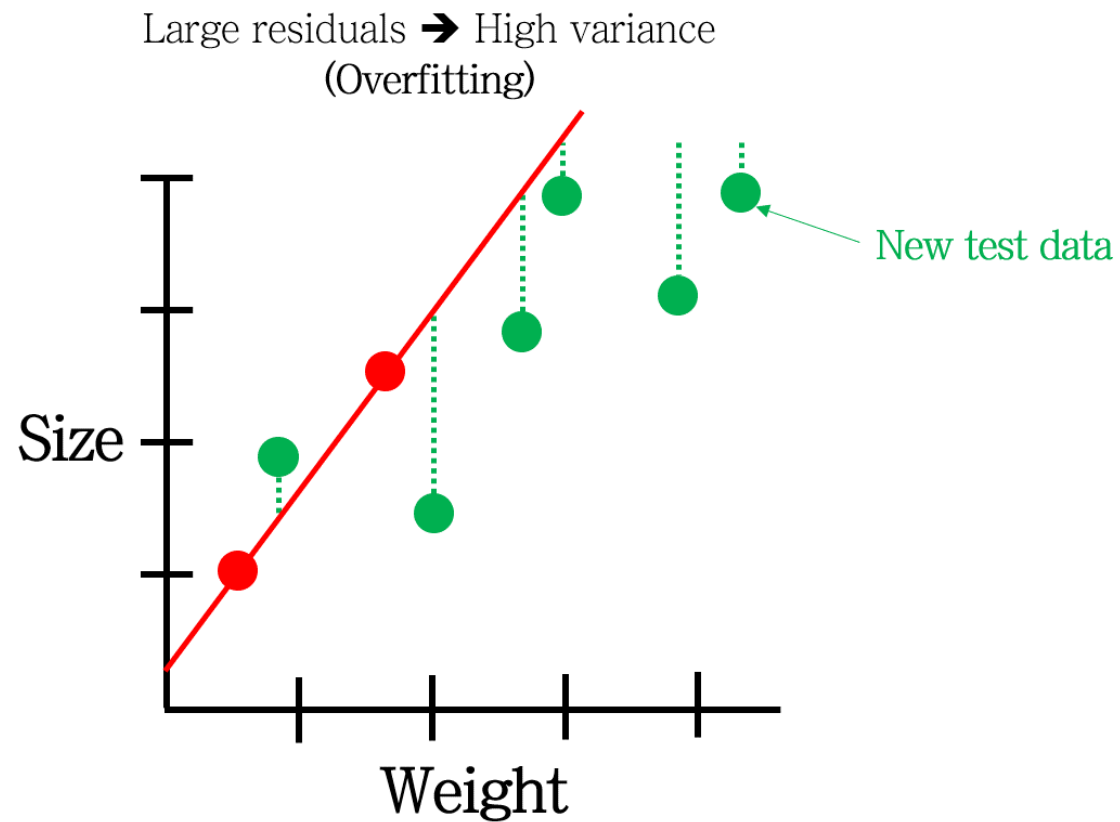
Fit a line with Least Squares



Linear Regression

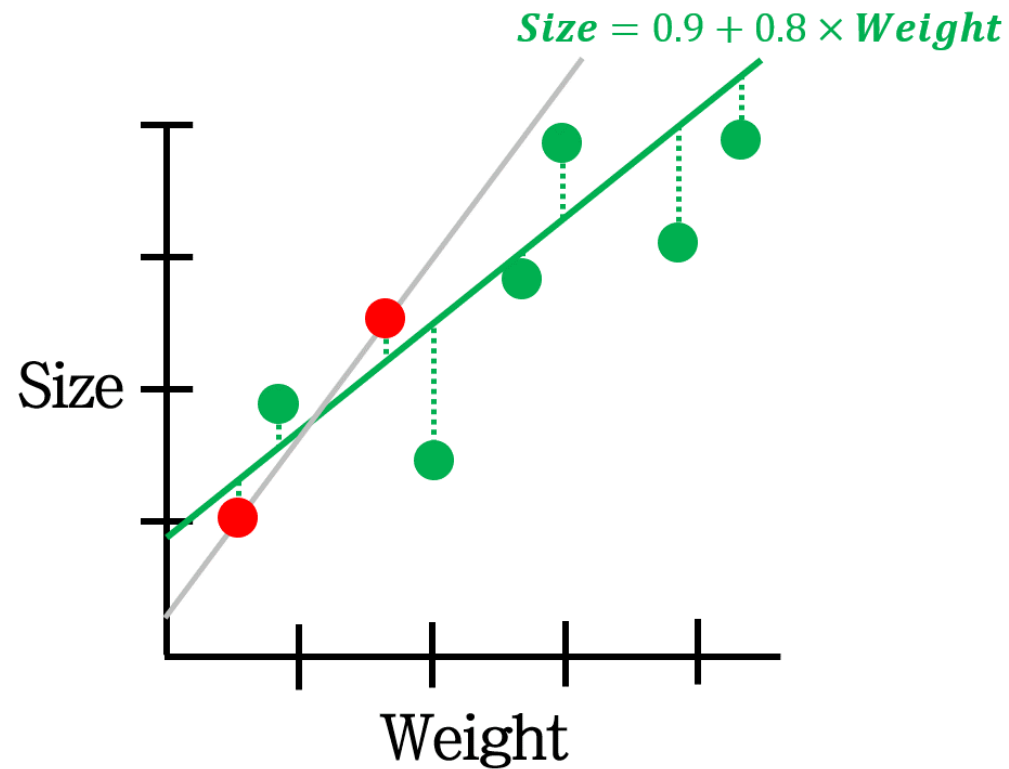


Overfitting

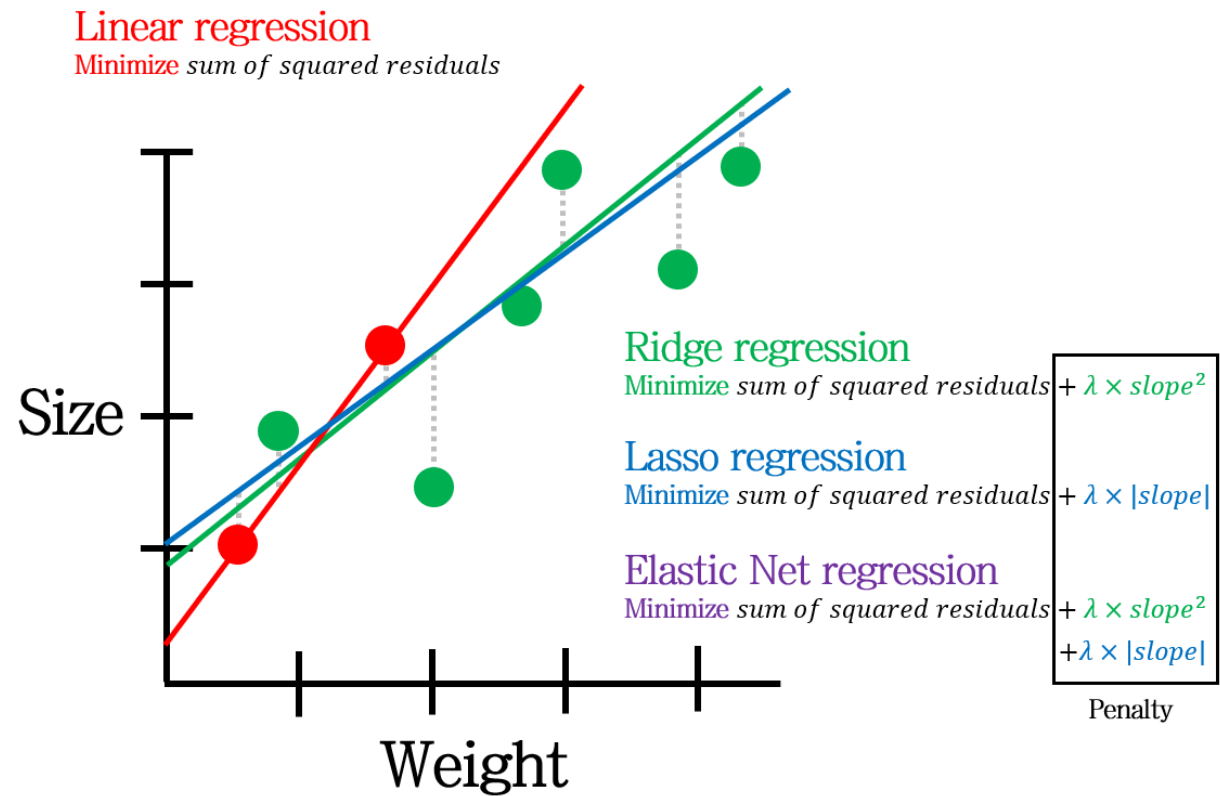


Regularization

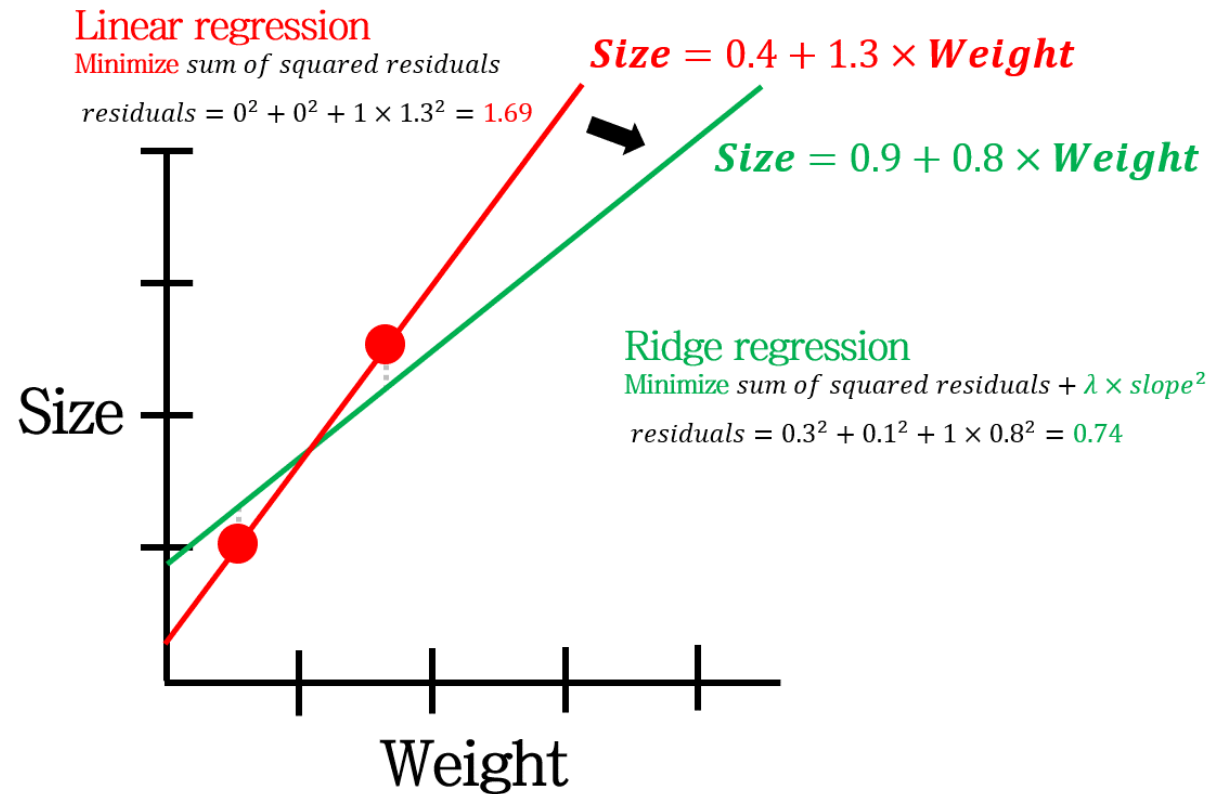
Increased small bias, but significantly less variance



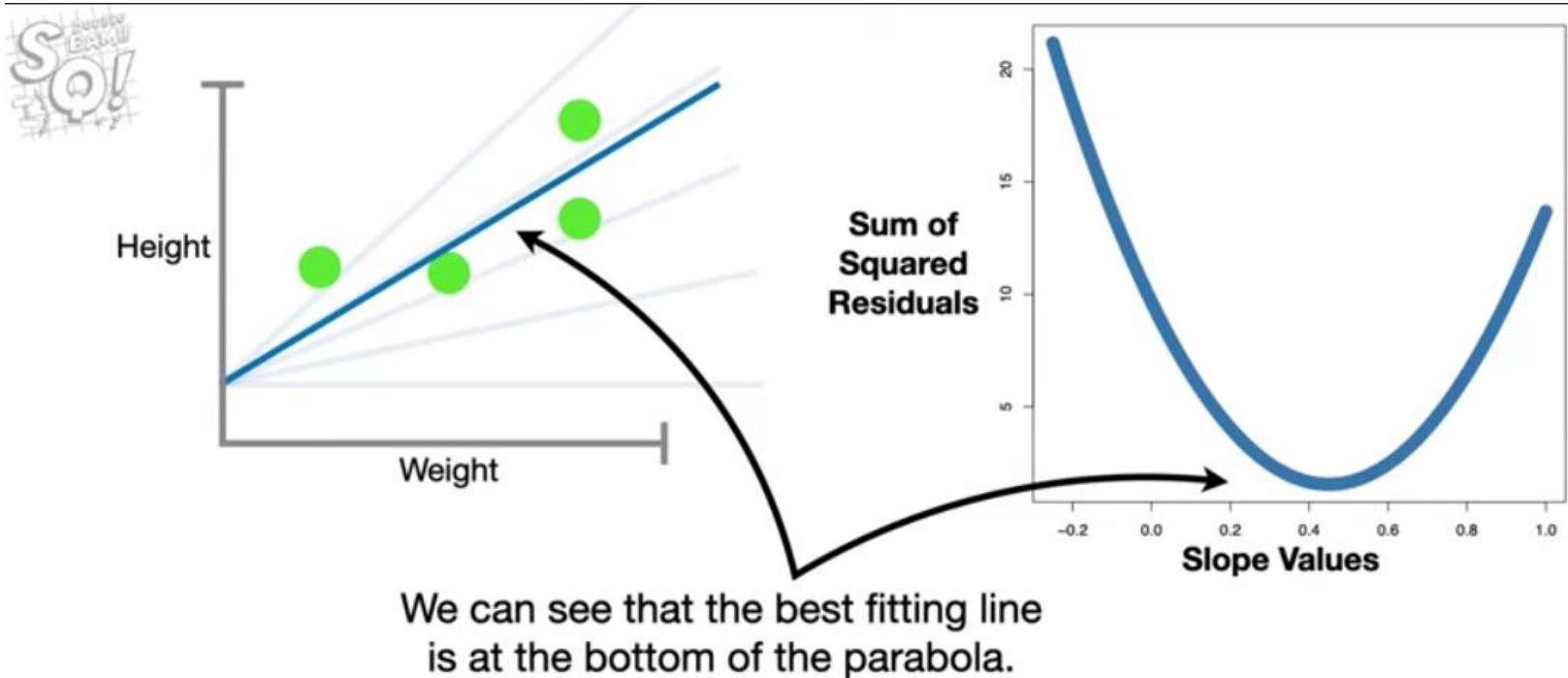
Regularization



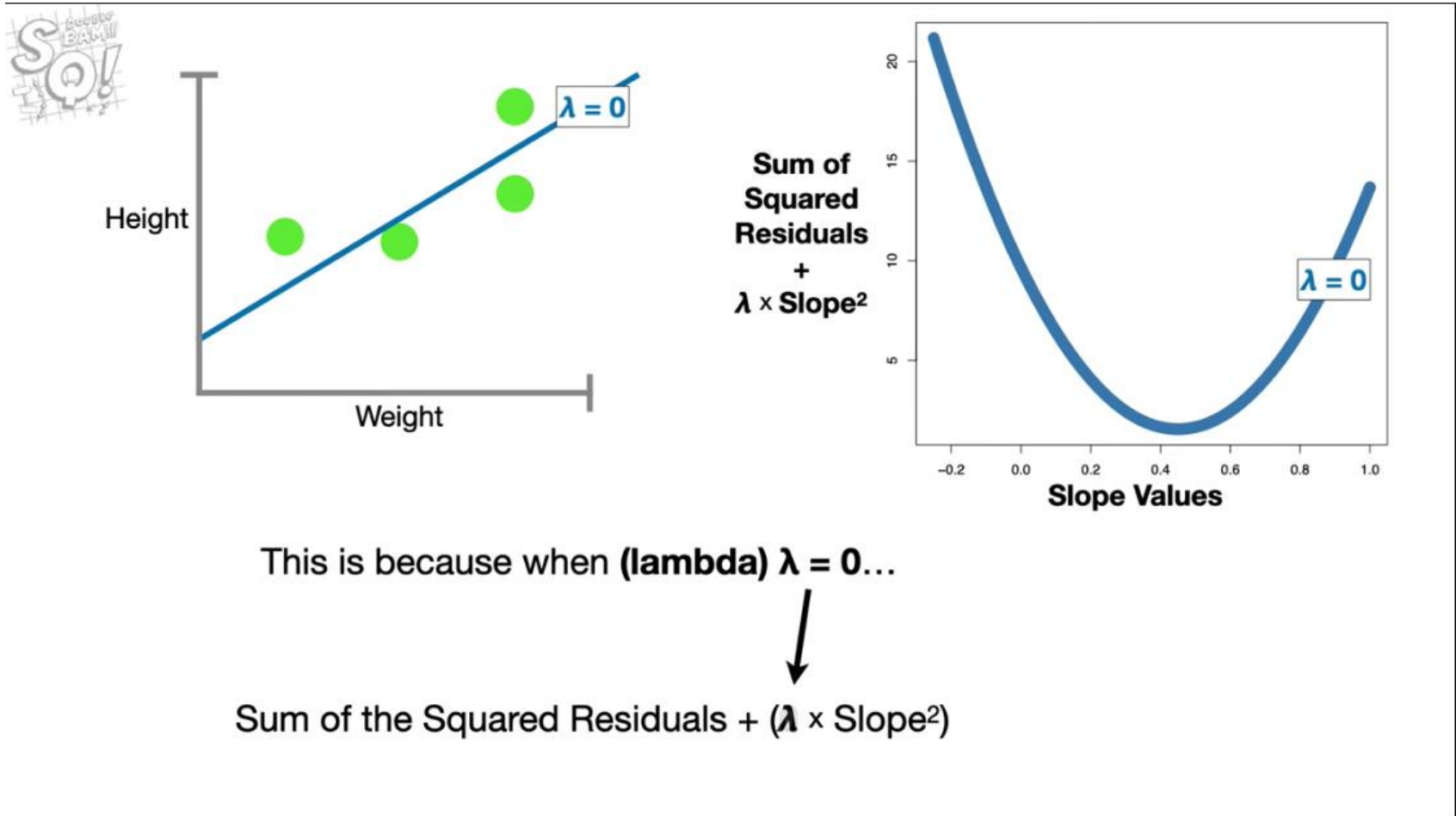
Ridge Regression



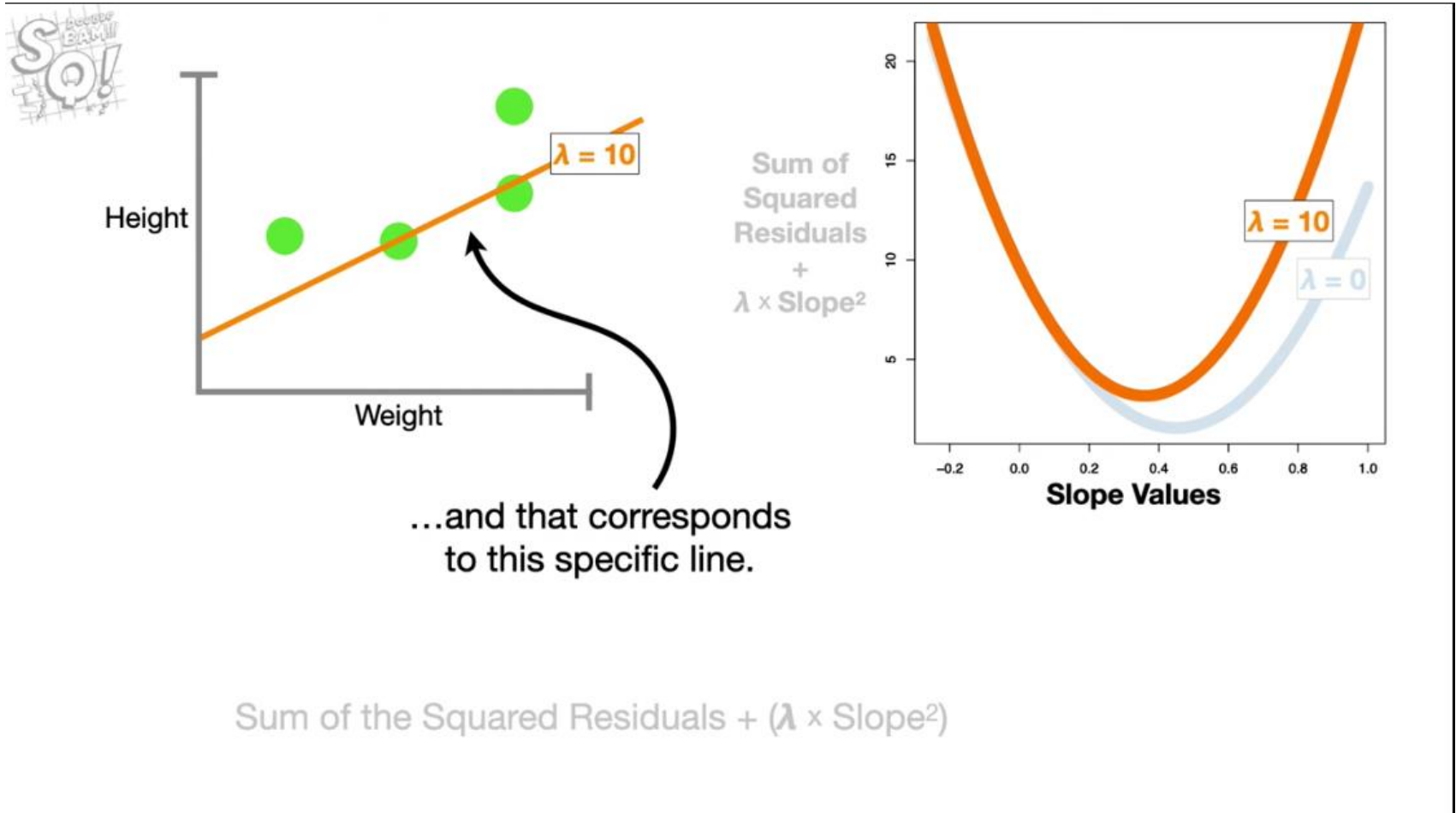
Linear Regression



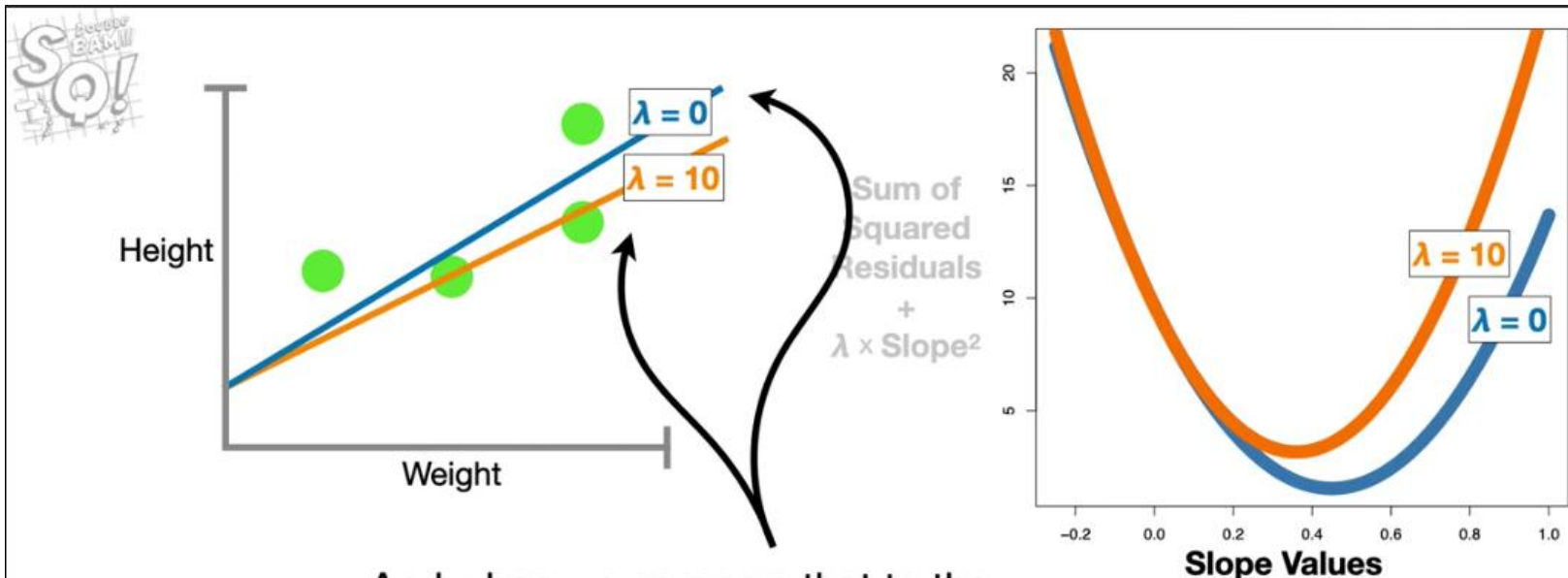
Ridge Regression



Ridge Regression



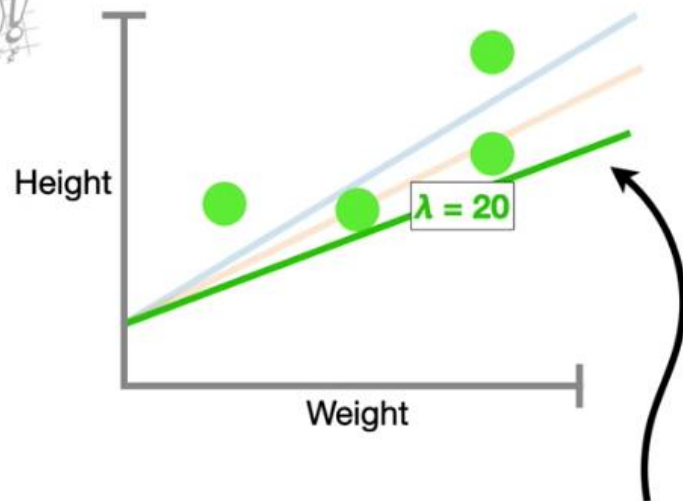
Ridge Regression



And when we compare that to the optimal slope when (**lambda**) $\lambda = 0$, we see that setting $\lambda = 10$ results in a smaller optimal slope.

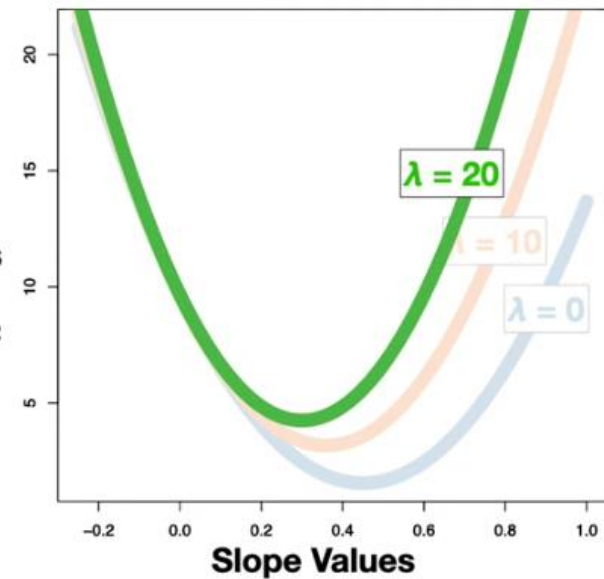
Sum of the Squared Residuals + ($\lambda \times \text{Slope}^2$)

Ridge Regression



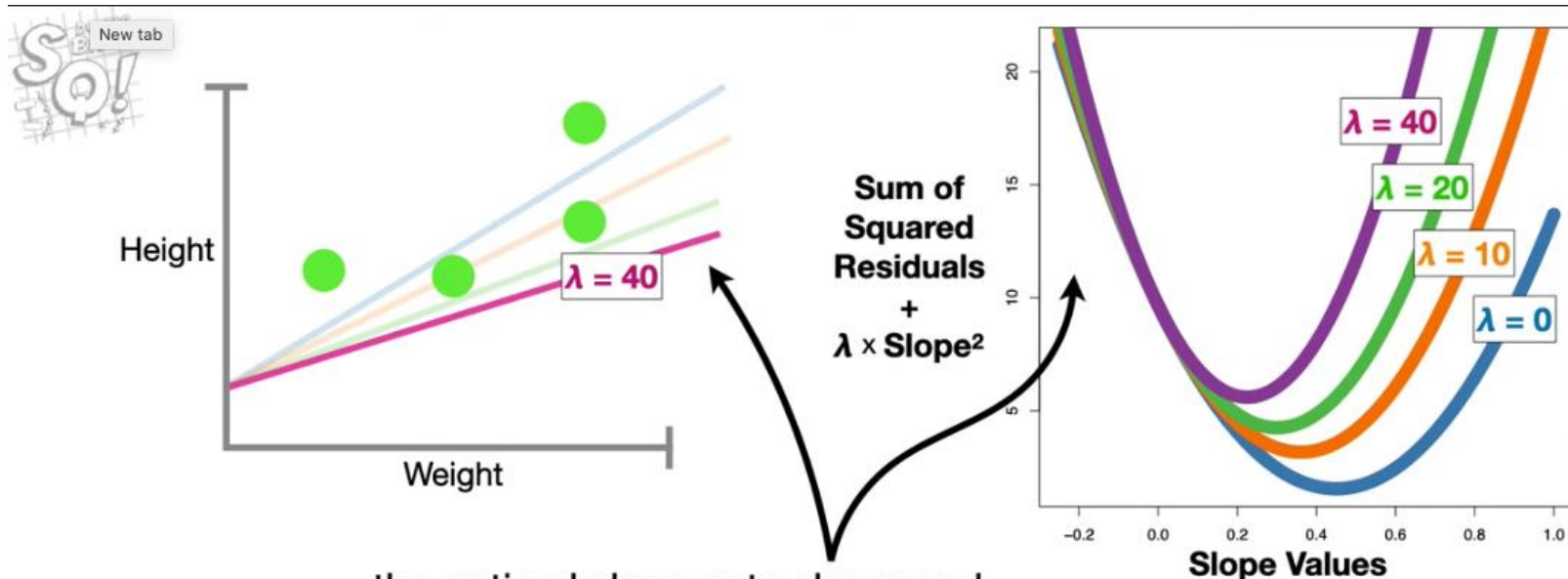
Sum of
Squared
Residuals
+
 $\lambda \times \text{Slope}^2$

...and the optimal slope has
shrunk some more.



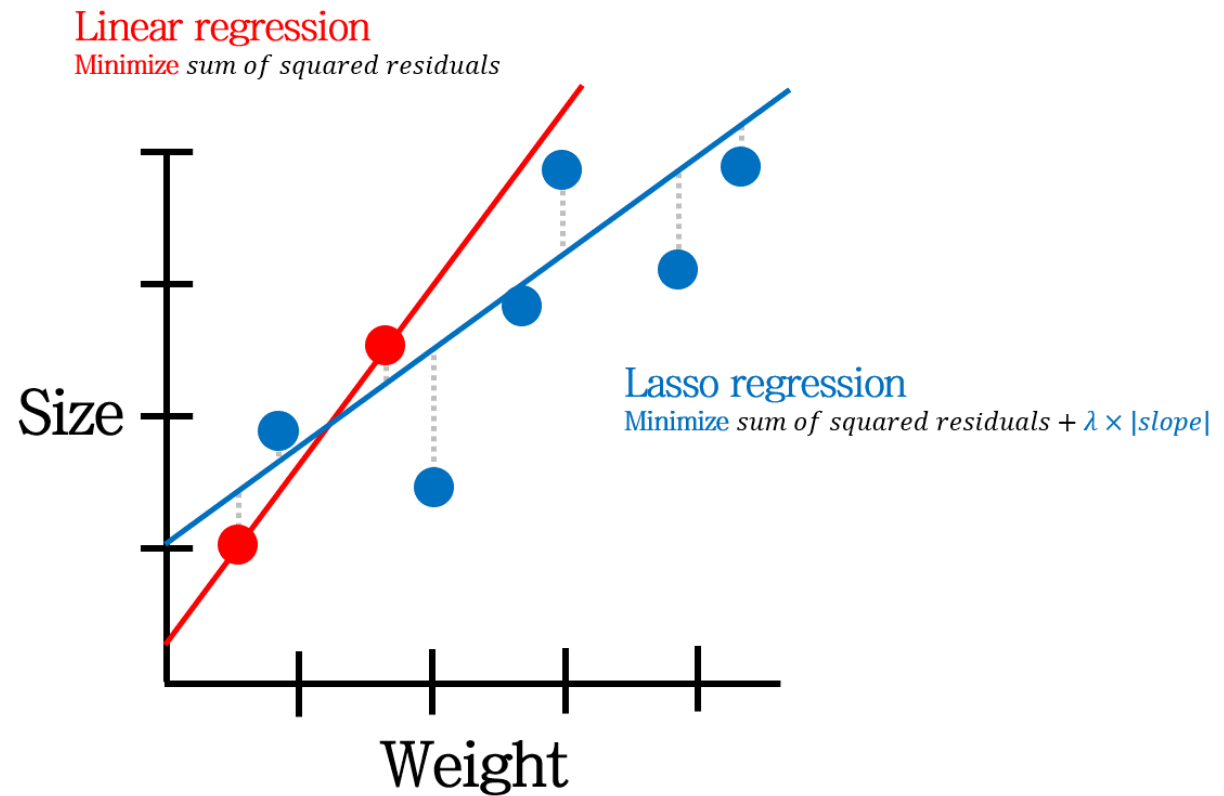
Sum of the Squared Residuals + $(\lambda \times \text{Slope}^2)$

Ridge Regression

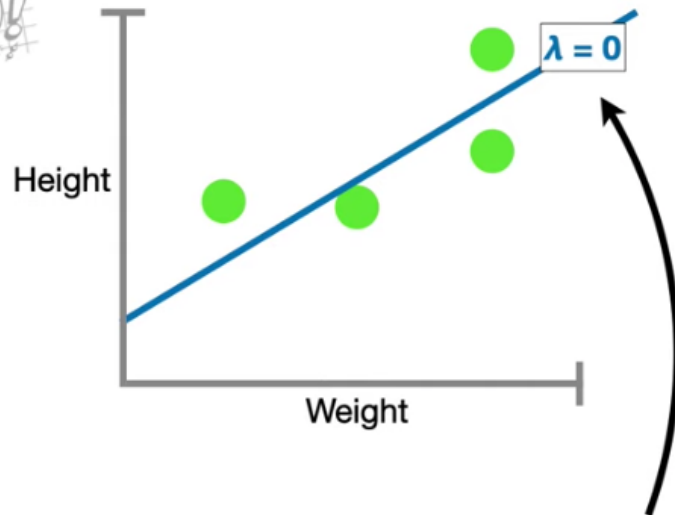


Sum of the Squared Residuals + ($\lambda \times \text{Slope}^2$)

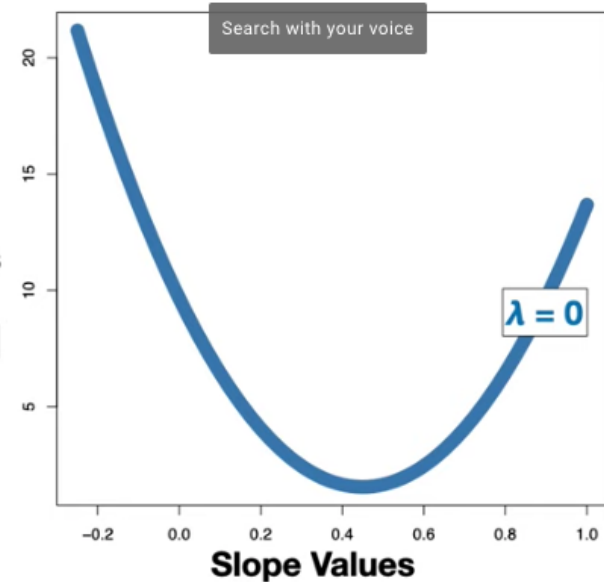
Lasso Regression



Lasso Regression

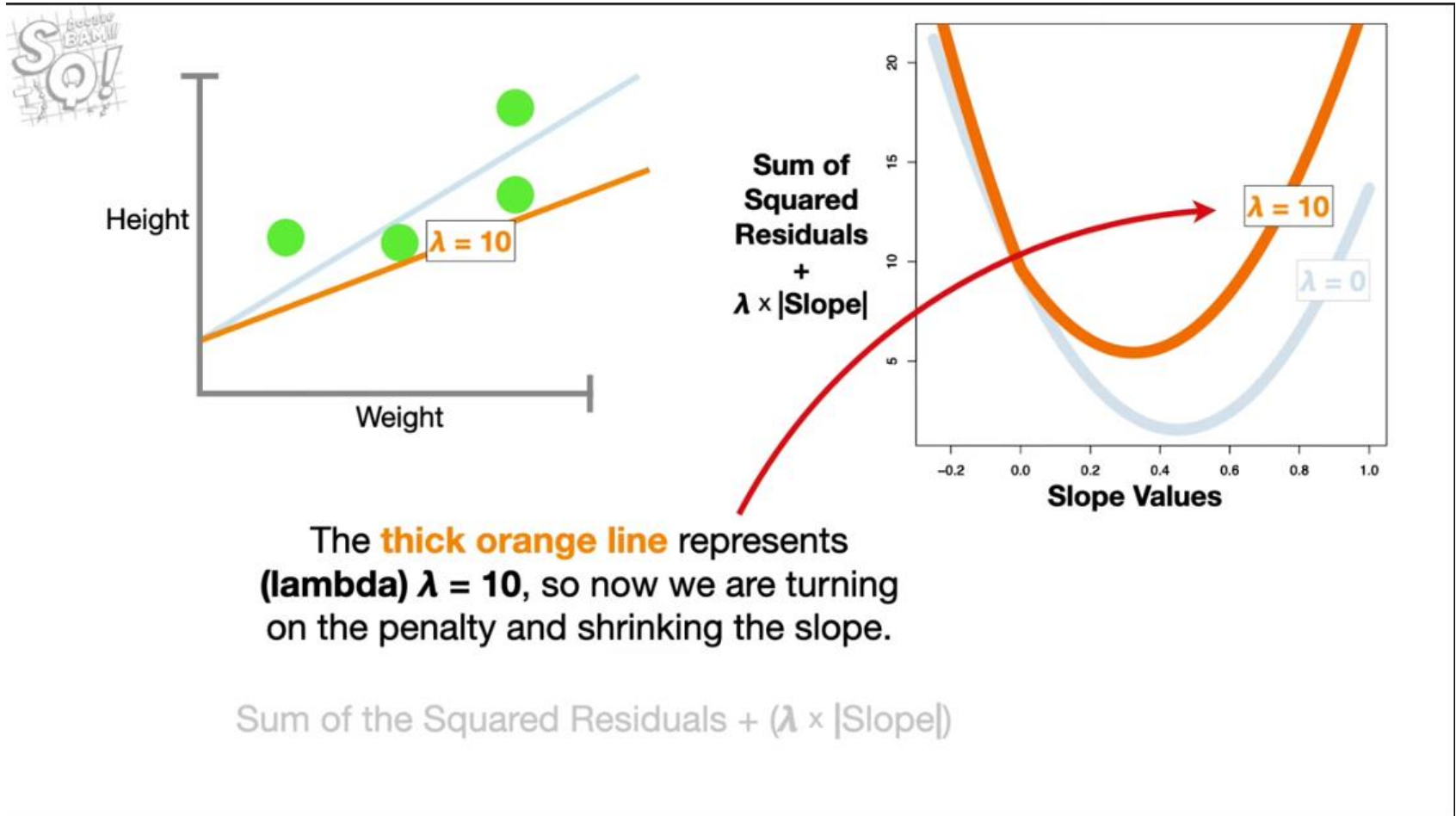


Sum of
Squared
Residuals
+
 $\lambda \times |\text{Slope}|$

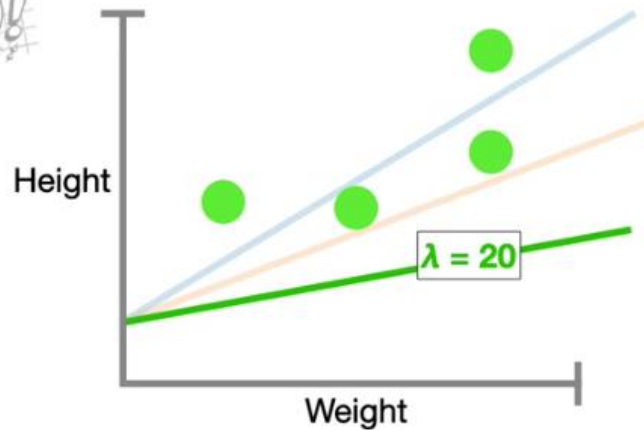


NOTE: Just like before, we'll keep track of the best fitting line, plus penalty, in this graph on the left.

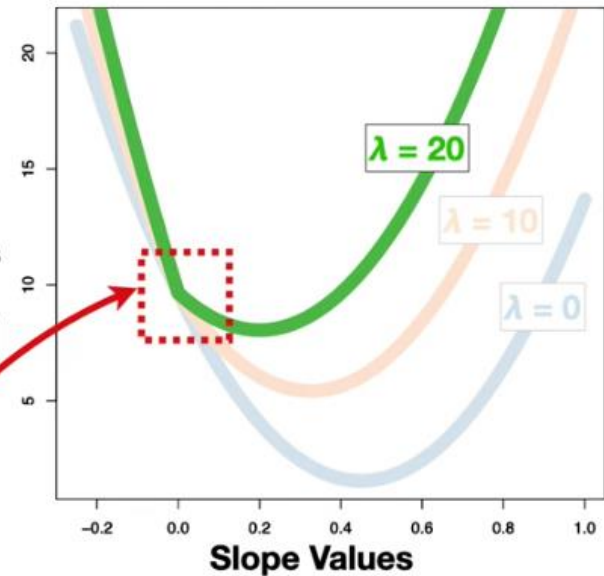
Lasso Regression



Lasso Regression



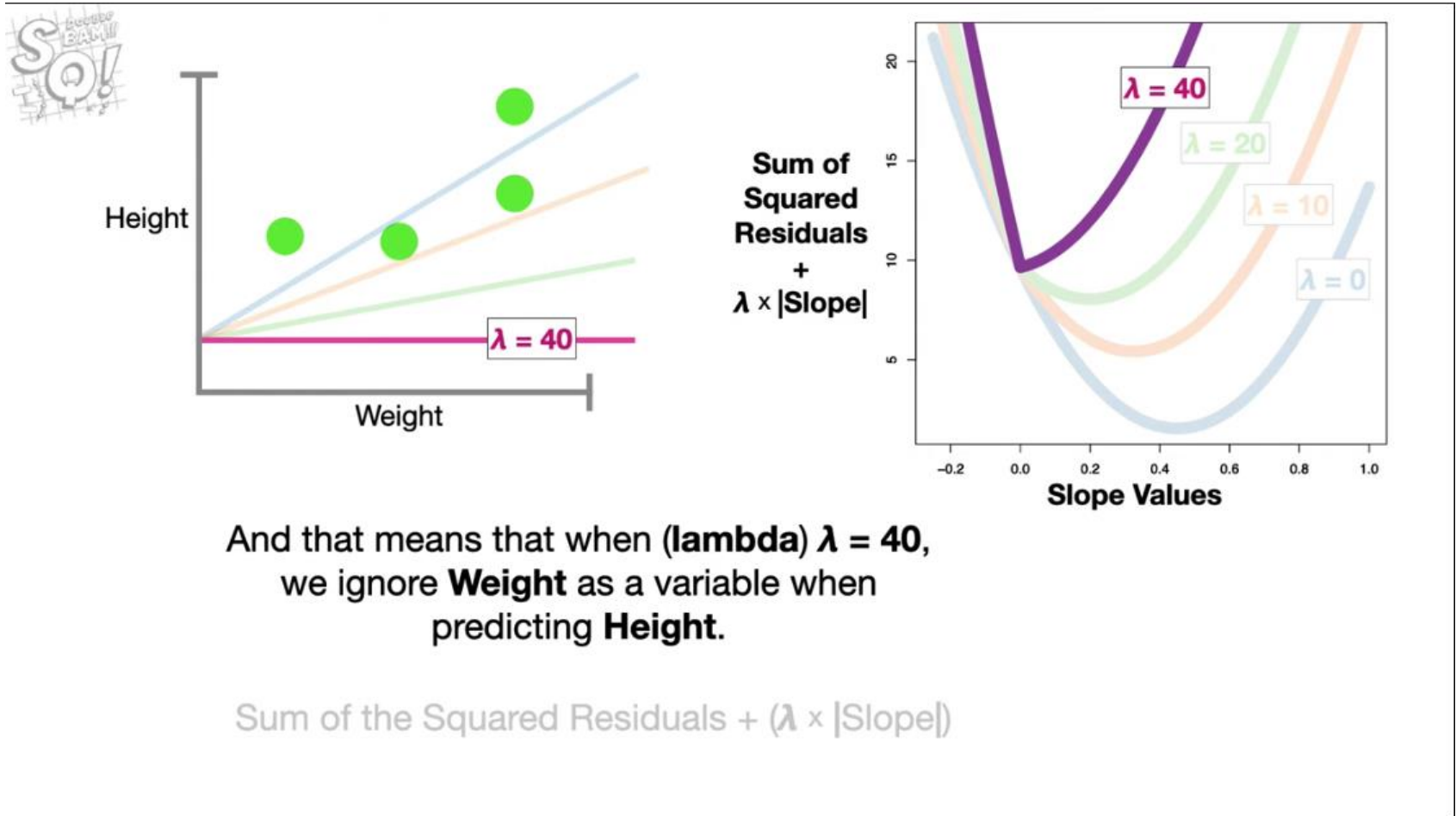
Sum of
Squared
Residuals
+
 $\lambda \times |\text{Slope}|$



...and this kink at **0** is becoming
more **prominent**.

Sum of the Squared Residuals + $(\lambda \times |\text{Slope}|)$

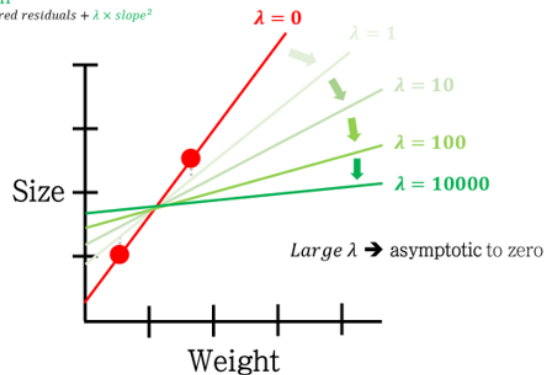
Lasso Regression



Ridge and Lasso Regression

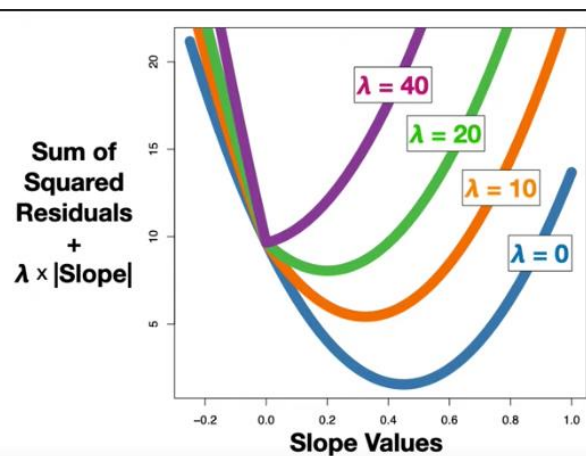
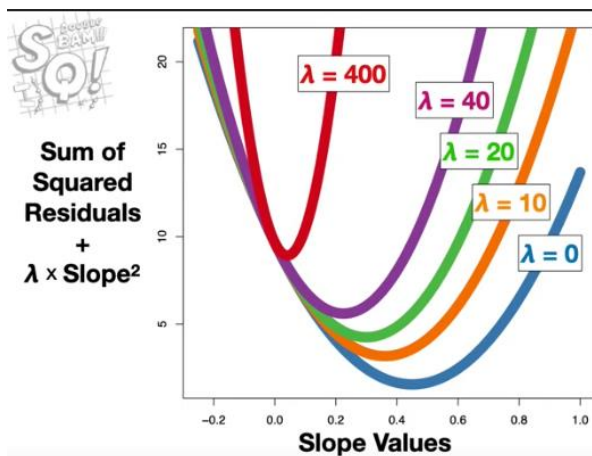
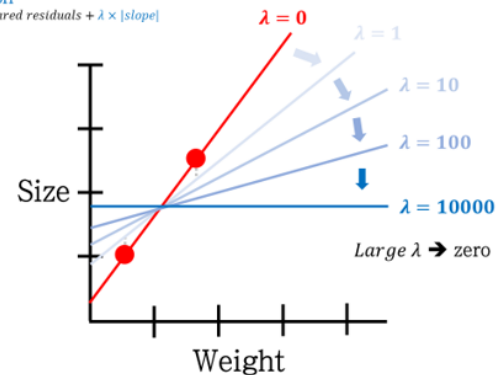
Ridge regression

Minimize sum of squared residuals + $\lambda \times \text{slope}^2$



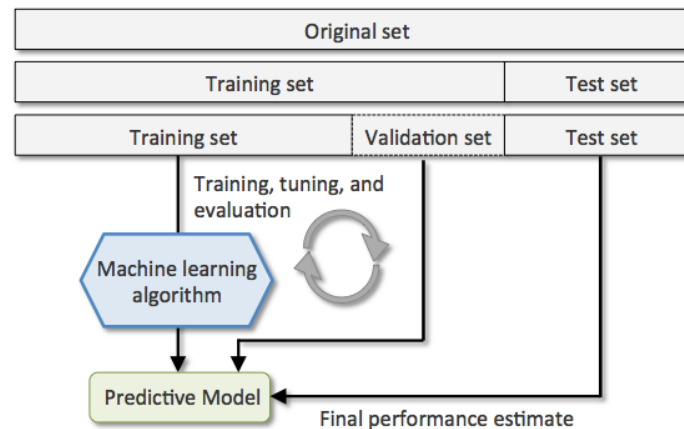
Lasso regression

Minimize sum of squared residuals + $\lambda \times |\text{slope}|$

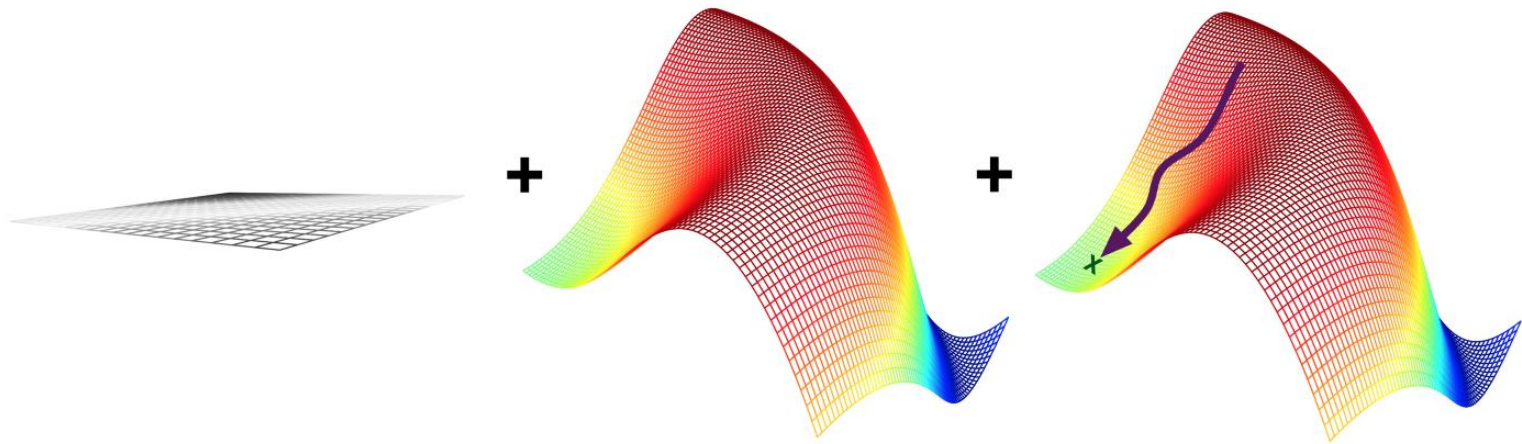


Data

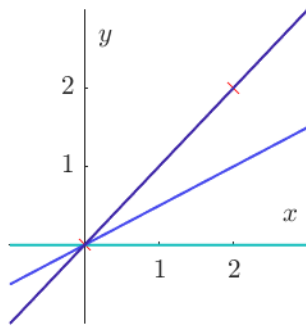
Data	Purpose	Example
Training	Optimize Model	$w_0, w_1, \dots$
Validation	Optimize Hyperparameter	Regularization Parameter
Testing Data	Performance Estimation	Reference Error



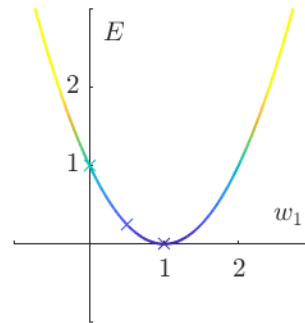
Machine Learning Components



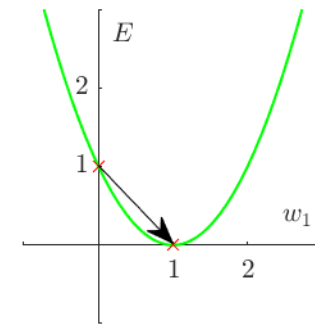
Representation



Evaluation



Optimization



Ridge Regression

Representation:

$$h(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_dx_d$$

Evaluation:

$$\frac{1}{2n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2n} \sum_{j=1}^d w_j^2$$

Optimization:

1. Normal equation
2. Gradient descent

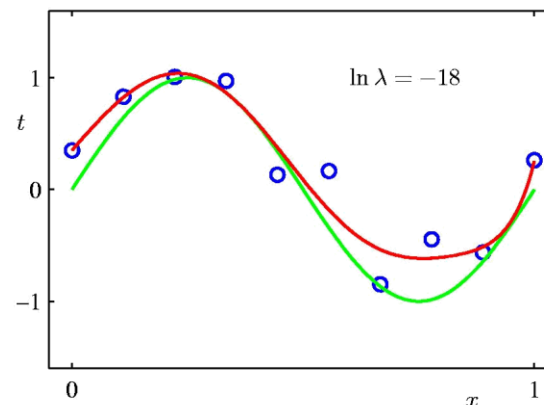
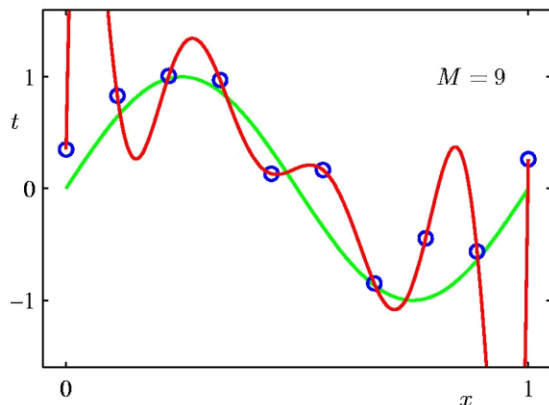
Source:

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Ridge Regression

Cost Function

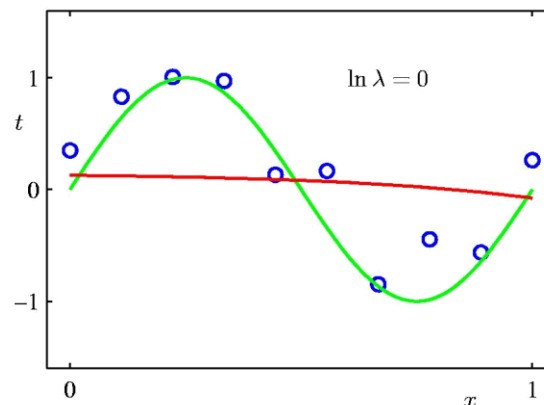
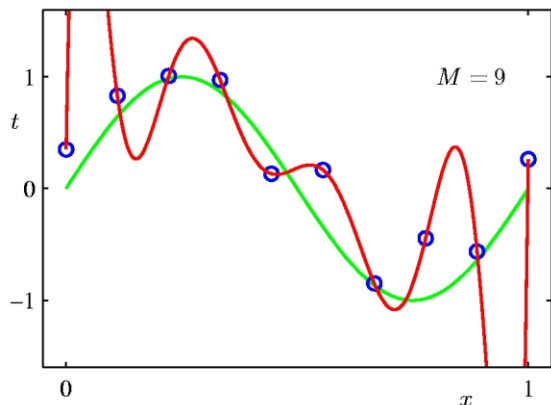
$$E(w) = \frac{1}{2n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2n} \sum_{j=1}^d w_j^2$$



Ridge Regression

Cost Function

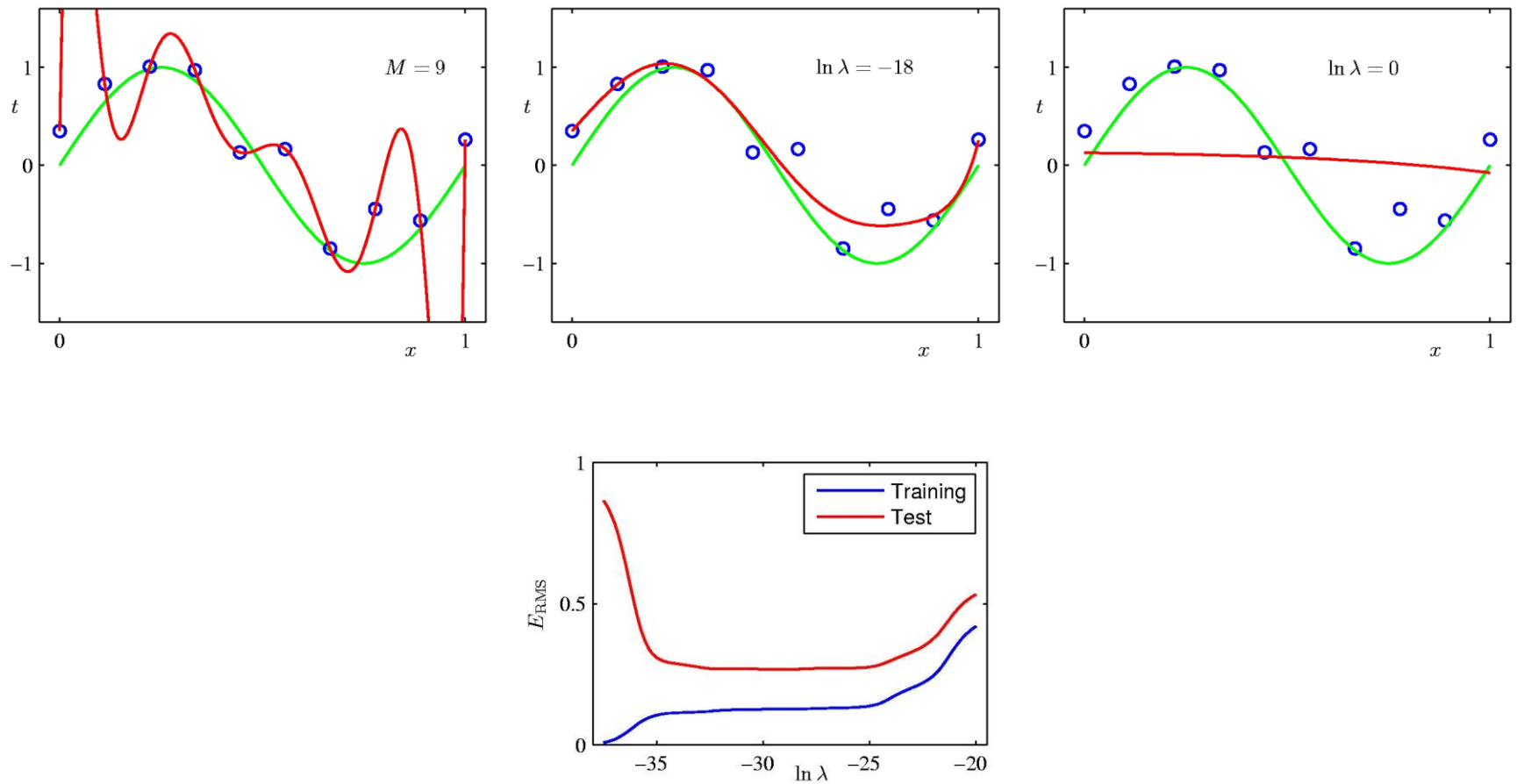
$$E(w) = \frac{1}{2n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2n} \sum_{j=1}^d w_j^2$$



Coefficient

$\ln(\lambda)$	$-\infty$	-18	0
w_0	0.35	0.35	0.13
w_1	232.37	4.74	-0.05
w_2	-5,321.83	-0.77	-0.06
w_3	48,568.31	-31.97	-0.05
w_4	-231,639.30	-3.89	-0.03
w_5	640,042.26	55.28	-0.02
w_6	-1,061,800.52	41.32	-0.01
w_7	1,042,400.18	-45.95	-0.00
w_8	-557,682.99	-91.53	0.00
w_9	125,201.43	76.28	0.01

Error



Ridge Regression

Cost Function

$$E(w) = \frac{1}{2n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2n} \sum_{j=1}^d w_j^2$$

Gradient

$$\frac{\partial}{\partial \theta_j} E(w) = \frac{1}{n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)}) x_j^{(i)} \text{ for } j = 0$$

$$\frac{\partial}{\partial \theta_j} E(w) = \frac{1}{n} \left(\sum_{i=1}^n (h(x^{(i)}) - y^{(i)}) x_j^{(i)} + \lambda w_j \right) \text{ for } j \geq 1$$

Gradient Descent

Repeat {

$$w_0 := w_0 - \alpha \frac{1}{n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$w_j := w_j - \alpha \left(\frac{1}{n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{n} w_j \right)$$

}

or

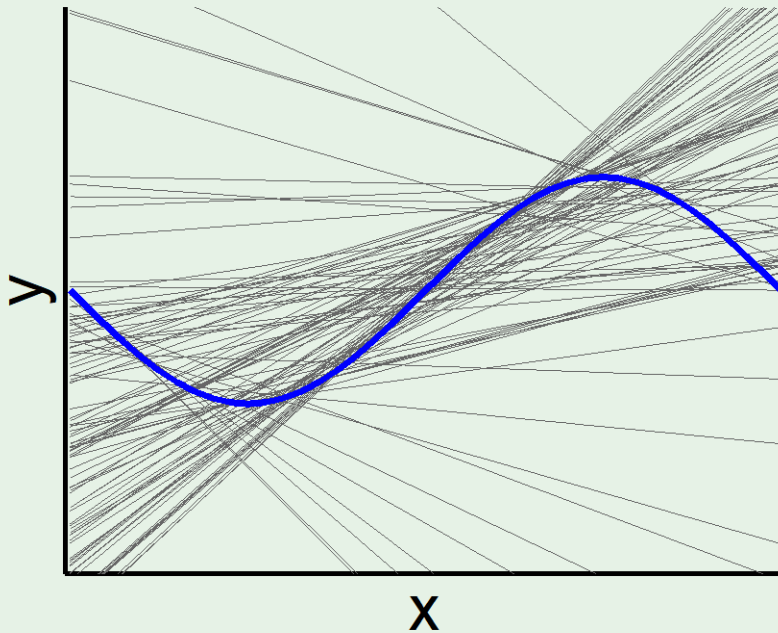
$$w_j := w_j \left(1 - \alpha \frac{\lambda}{n} \right) - \alpha \frac{1}{n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

Normal Equation

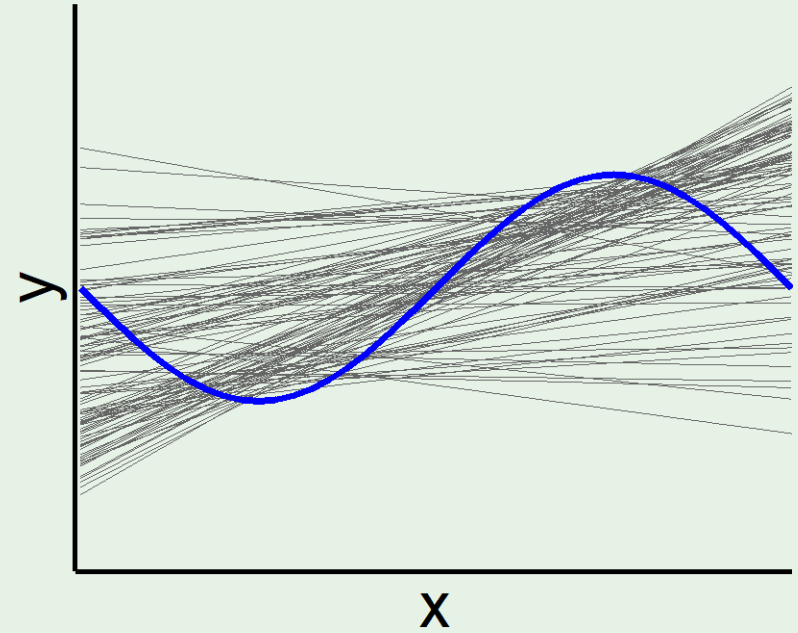
$$w = \left(X^T X + \lambda \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} \right)^{-1} X^T y$$

กรณีศึกษา

สมมติว่าฟังก์ชันเป้าหมาย คือ $\sin(\pi x)$ ซักตัวอย่าง x ด้วยการแจกแจงแบบเอกรูป (Uniform) ในช่วง $[-1, 1]$ ออกมา 2 ตัวอย่าง สร้างแบบจำลองด้วยการสมการเชิงเส้นที่มีและไม่มีทำให้เป็นปกติ



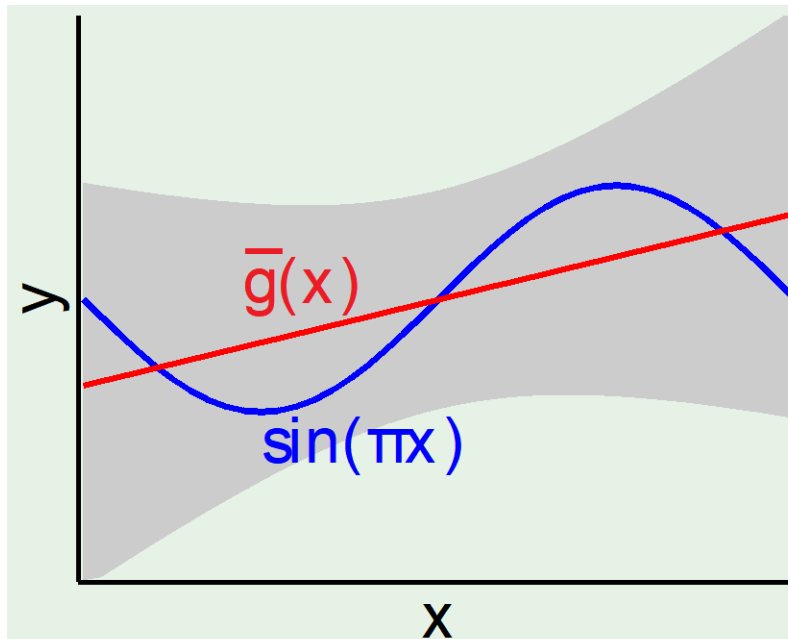
without regularization



with regularization

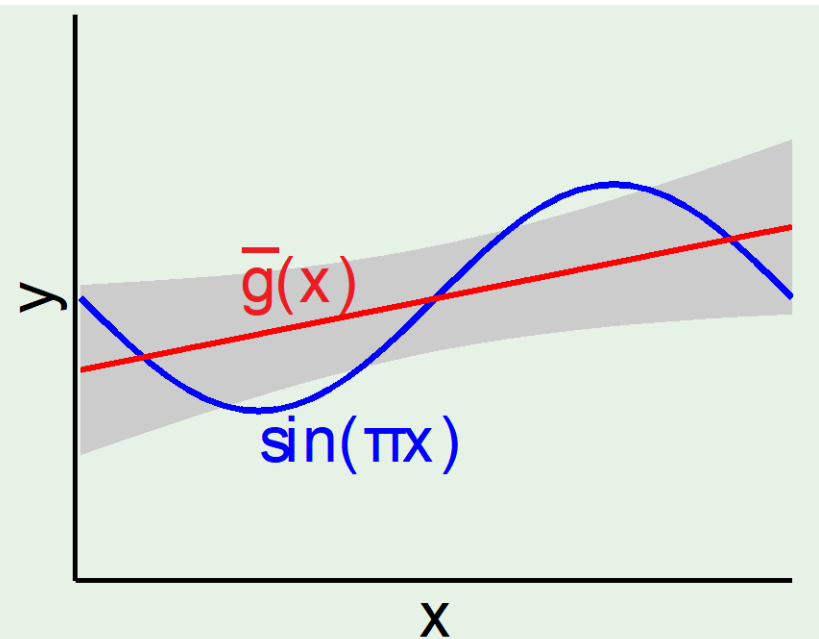
กรณีศึกษา

ในตัวอย่างก่อนหน้านี้แบบจำลองค่าคงที่จะมี $E_{out} = 0.75$ และแบบจำลองเชิงเส้นที่ไม่มีการทำให้เป็นปรกติที่มี $E_{out} = 1.90$ แต่แบบจำลองเชิงเส้นที่มีการทำให้เป็นปรกติจะมี $E_{out} = 0.56$ จะเป็นแบบจำลองที่ดีที่สุด



bias = 0.21

var = 1.69



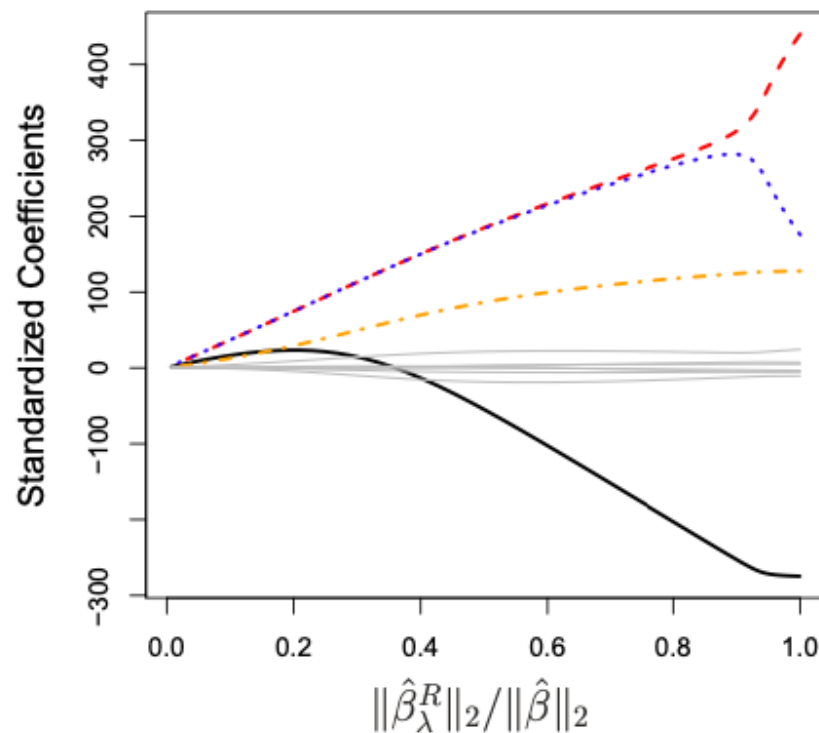
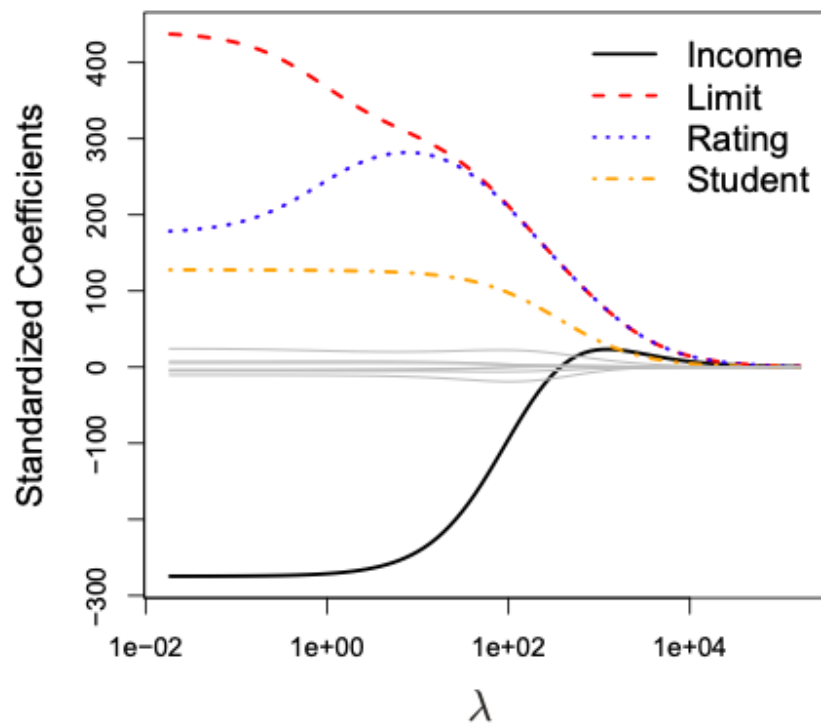
bias = 0.23

var = 0.33

Regularization

จะลดความแปรปรวนของแบบจำลองเชิงเส้นจาก 1.69 เหลือเพียง 0.33 โดยการยอมให้มีความเอนเอียงเพิ่มขึ้นจาก 0.21 เป็น 0.23 ส่งผลให้ลดค่าคลาดเคลื่อนนอกตัวอย่างซึ่งเกิดจากผลรวมของความเอนเอียงและความแปรปรวน

Ridge Regression



The Bias-Variance tradeoff

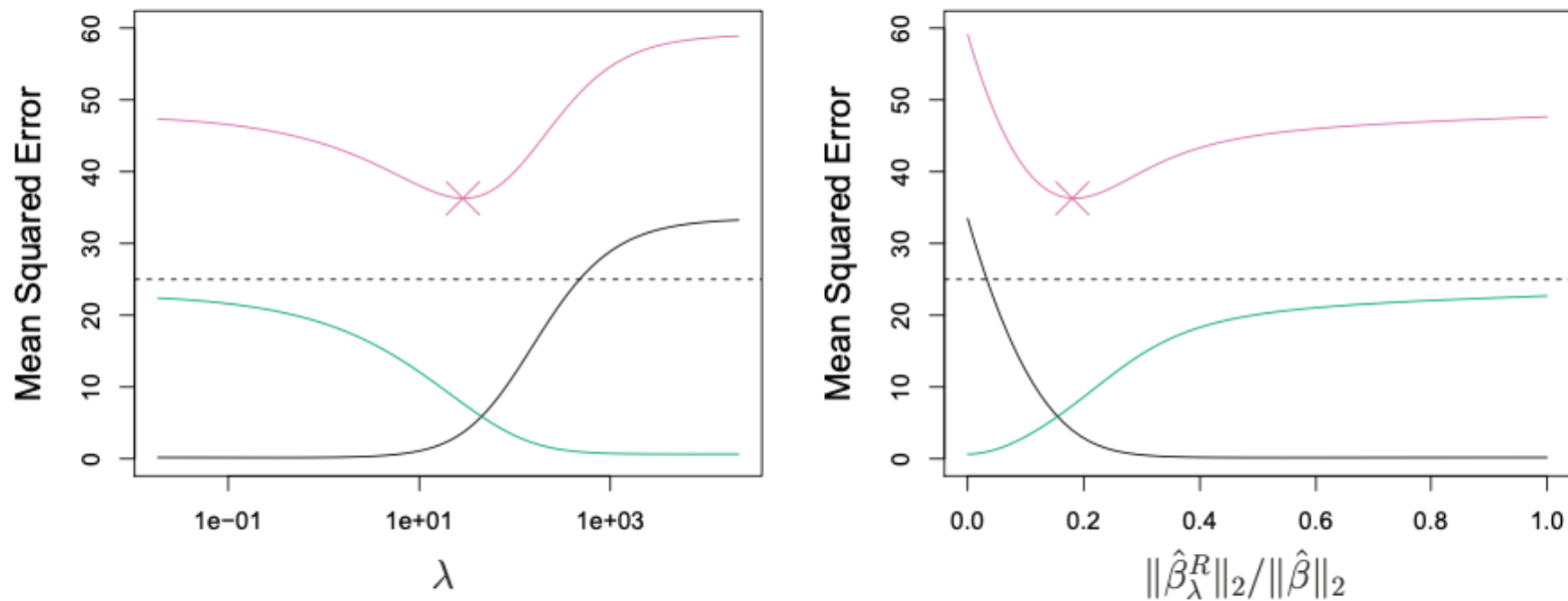


FIGURE 6.5. Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set, as a function of λ and $\|\hat{\beta}_\lambda^R\|_2^2 / \|\hat{\beta}\|_2^2$. The horizontal dashed lines indicate the minimum possible MSE. The purple crosses indicate the ridge regression models for which the MSE is smallest.

Regularization

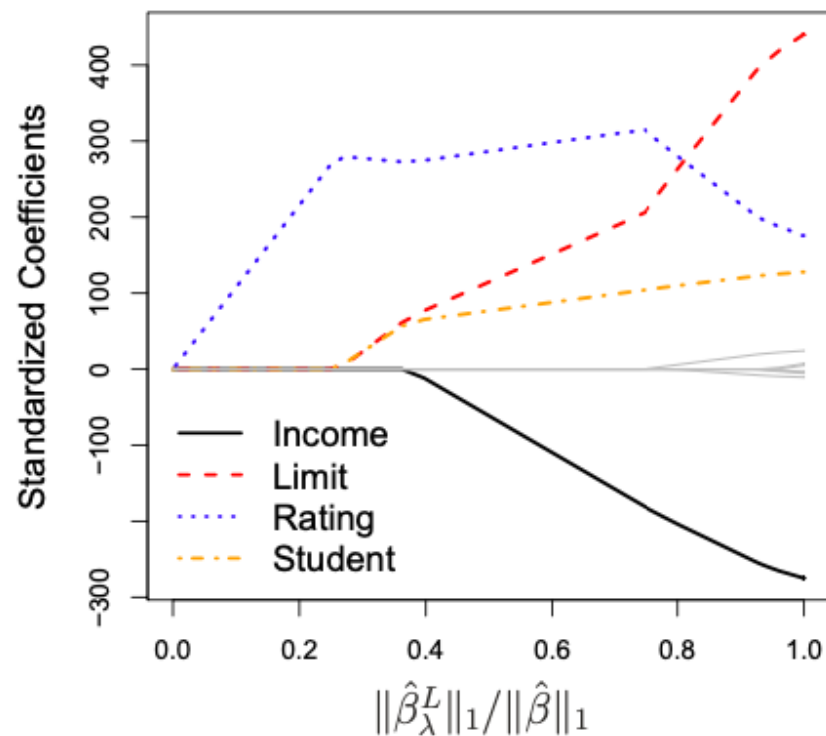
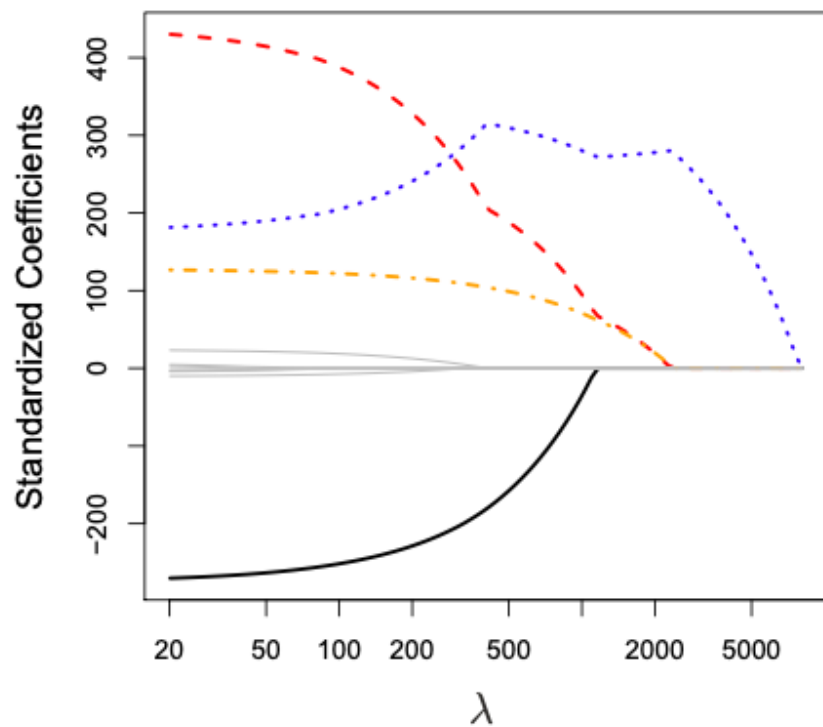
Ridge Regression

$$E(w) = \frac{1}{2n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2n} \sum_{j=1}^d w_j^2$$

Lasso

$$E(w) = \frac{1}{2n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2n} \sum_{j=1}^d |w_j|$$

Lasso



The Variable Selection Property of the Lasso

One can show that the lasso and ridge regression coefficient estimates solve the problems

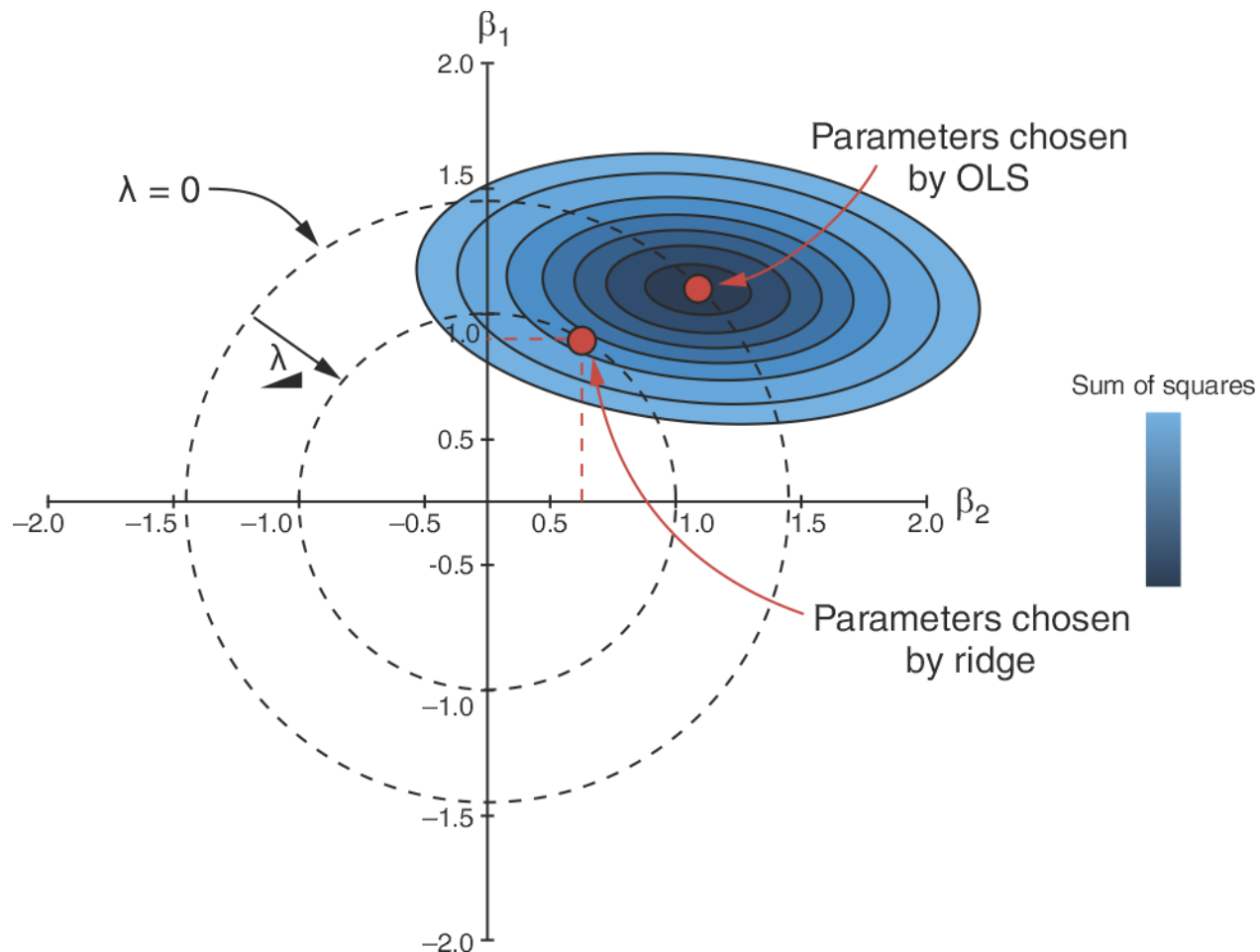
$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s \quad (6.8)$$

and

$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s, \quad (6.9)$$

Ridge Regression

$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s,$$



Constraint optimization

กำหนดให้ $\mathbf{w}_{\text{lin}}$ คือ สัมประสิทธิ์ของแบบจำลองที่ได้จากการถดถอยเชิงเส้น และสามารถหาได้จาก

$$\mathbf{w}_{\text{lin}} = \arg \min_{\mathbf{w}} E_{\text{in}}(\mathbf{w})$$

และการทำให้เป็นปรกติทำได้โดยใส่เงื่อนไขบังคับเพิ่มเติมเข้าไปด้วยสมการ

$$\sum_{j=0}^p w_j^2 = \mathbf{w}^T \mathbf{w} \leq C$$

และกำหนดให้ผลลัพธ์ที่ได้คือ $\mathbf{w}_{\text{reg}}$ หรือสัมประสิทธิ์ของแบบจำลองที่ได้จากการถดถอยเชิงเส้นที่มีการทำให้เป็นปรกติ

Constraint optimization

พิจารณาการหาผลลัพธ์ของ $\mathbf{w}_{\text{reg}}$ ถ้าความซับซ้อนของแบบจำลองเชิงเส้นหรือผลรวมของค่าสัมประสิทธิ์ของแบบจำลองยกกำลังสองมีค่าน้อยกว่าค่าพารามิเตอร์ C หรือ $\mathbf{w}_{\text{lin}}^T \mathbf{w}_{\text{lin}} \leq C$ แล้วแบบจำลองเชิงเส้นที่ทำให้เป็นปรกติจะเป็นแบบจำลองการถดถอยเชิงเส้นหรือ $\mathbf{w}_{\text{reg}} = \mathbf{w}_{\text{lin}}$ เนื่องจาก $\mathbf{w}_{\text{lin}} \leq \mathbf{w}_{\text{reg}}$

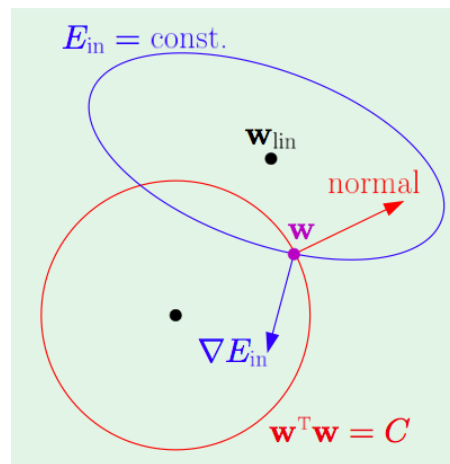
แต่ถ้าผลรวมของค่าสัมประสิทธิ์ของแบบจำลองยกกำลังสองมีค่ามากกว่าค่าพารามิเตอร์ C หรือ $\mathbf{w}_{\text{lin}}^T \mathbf{w}_{\text{lin}} > C$ แล้วแบบจำลองเชิงเส้นที่ทำให้เป็นปรกติจะสามารถหาได้จาก $\mathbf{w}_{\text{reg}}^T \mathbf{w}_{\text{reg}} = C$

Constraint optimization

ดังนั้นปัญหาการหาค่า $\mathbf{w}_{\text{reg}}$ ที่ทำให้ค่าคลาดเคลื่อนในตัวอย่างน้อยสุด จะมีเงื่อนไขบังคับเปลี่ยนเป็น $\mathbf{w}^T \mathbf{w} = C$ ดังรูป ค่าสัมประสิทธิ์ของแบบจำลองจะต้องอยู่บนวงกลมสีแดง และเวกเตอร์ที่ตั้งฉากกับเส้นสัมผัสคือ เวกเตอร์ปกติ (Normal Vector)

คอนทัวร์ของค่าคลาดเคลื่อนในตัวอย่างที่คงที่ค่าหนึ่งแสดงด้วยเส้นสีน้ำเงิน เวกเตอร์ปกติของคอนทัวร์นี้คือ เวกเตอร์ความชัน $\nabla E_{\text{in}}(\mathbf{w})$

จากรูป พบว่าเวกเตอร์ทั้งสองไม่ขนานกัน ดังนั้น $\mathbf{w}$ จะไม่ใช่ค่าที่ดีที่สุดเนื่องจากการปรับค่า $\mathbf{w}$ จะทำให้ค่า $E_{\text{in}}(\mathbf{w})$ ลดลงได้



Constraint optimization

ดังนั้น ถ้า $\mathbf{w}_{\text{reg}}$ คือ ค่าที่เหมาะสมที่สุดแล้วจะมีพารามิเตอร์ λ ที่ทำให้

$$\nabla E_{\text{in}}(\mathbf{w}_{\text{reg}}) = -2 \frac{\lambda}{n} \mathbf{w}_{\text{reg}}$$

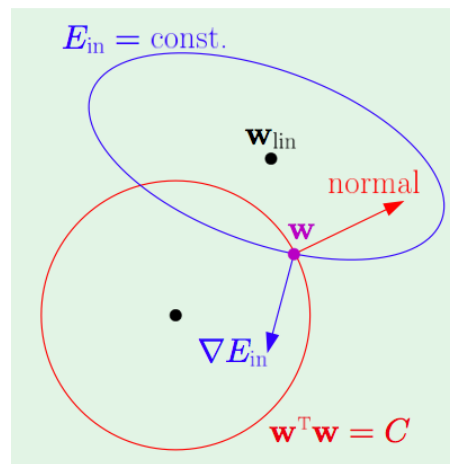
หรือ

$$\nabla(E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{n} \mathbf{w}^T \mathbf{w}) = 0$$

เนื่องจาก $\nabla(\mathbf{w}^T \mathbf{w}) = 2\mathbf{w}$ ดังนั้น $\mathbf{w}_{\text{reg}}$ จะหาได้จากการหาค่าคลาดเคลื่อนแต่งเติม (Augmented Error) น้อย

สุด

$$E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{n} \mathbf{w}^T \mathbf{w}$$



Solving for $\mathbf{w}_{\text{reg}}$

Minimize $E_{\text{in}}(\mathbf{w}) = \frac{1}{N} (\mathbf{Z}\mathbf{w} - \mathbf{y})^\top (\mathbf{Z}\mathbf{w} - \mathbf{y})$

subject to: $\mathbf{w}^\top \mathbf{w} \leq C$

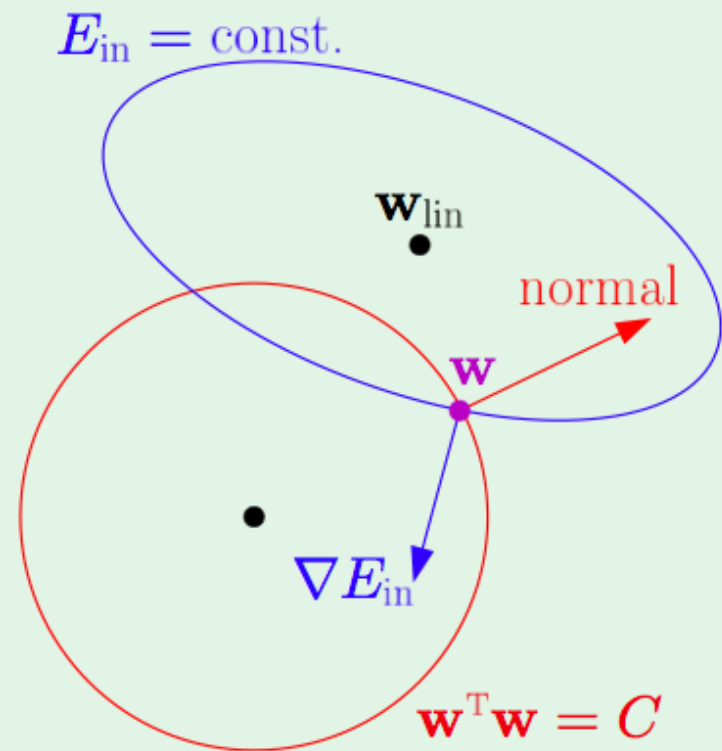
$$\nabla E_{\text{in}}(\mathbf{w}_{\text{reg}}) \propto -\mathbf{w}_{\text{reg}}$$

$$= -2\frac{\lambda}{N}\mathbf{w}_{\text{reg}}$$

$$\nabla E_{\text{in}}(\mathbf{w}_{\text{reg}}) + 2\frac{\lambda}{N}\mathbf{w}_{\text{reg}} = \mathbf{0}$$

Minimize $E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\mathbf{w}^\top \mathbf{w}$

$$\boxed{C \uparrow \quad \lambda \downarrow}$$



The solution

Minimize $E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N} \mathbf{w}^T \mathbf{w}$

$$= \frac{1}{N} \left((\mathbf{Z}\mathbf{w} - \mathbf{y})^T (\mathbf{Z}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \right)$$

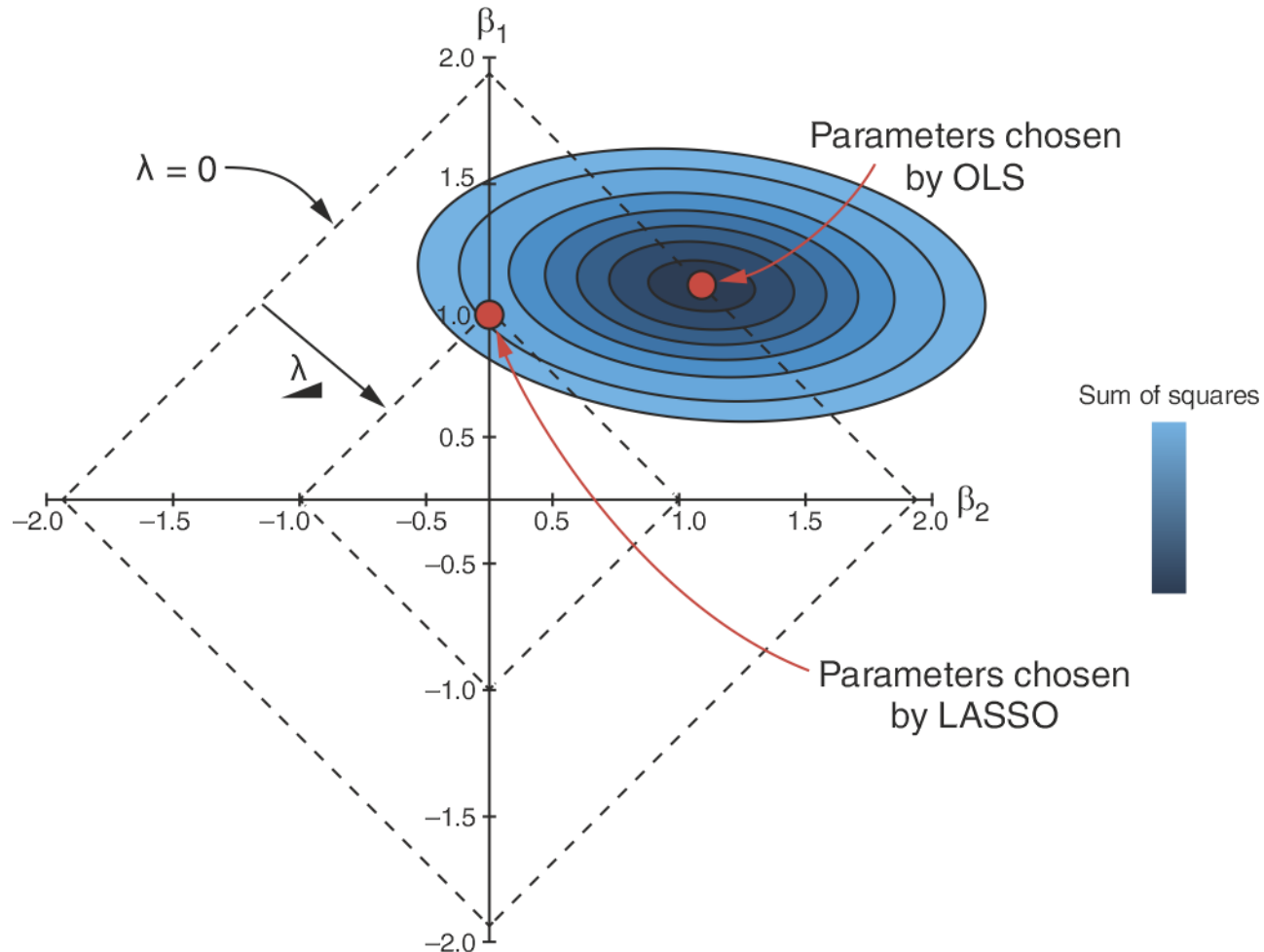
$$\nabla E_{\text{aug}}(\mathbf{w}) = \mathbf{0} \quad \implies \quad \mathbf{Z}^T (\mathbf{Z}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w} = \mathbf{0}$$

$$\boxed{\mathbf{w}_{\text{reg}} = (\mathbf{Z}^T \mathbf{Z} + \lambda \mathbf{I})^{-1} \mathbf{Z}^T \mathbf{y}} \quad (\text{with regularization})$$

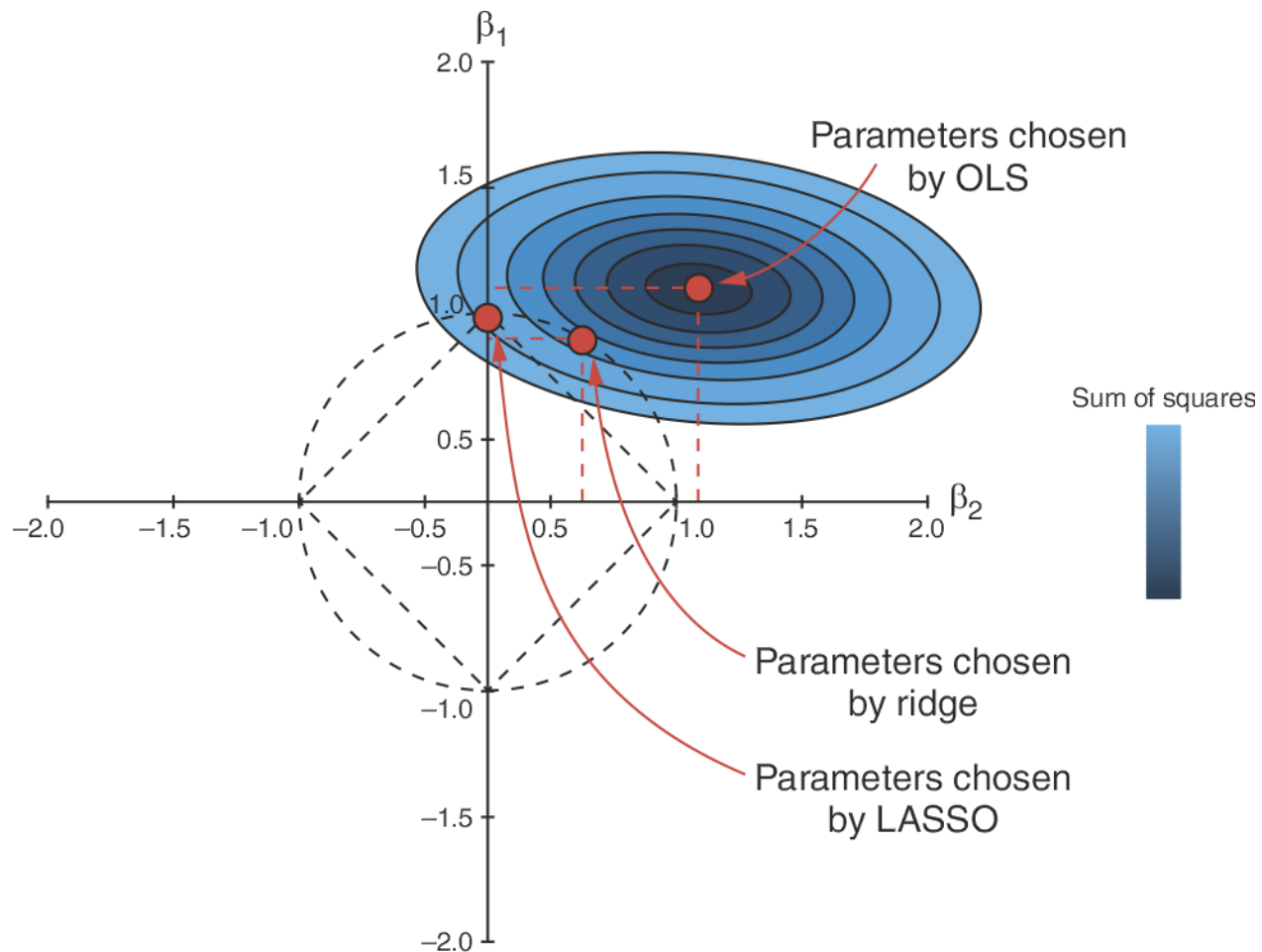
as opposed to $\mathbf{w}_{\text{lin}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}$ (without regularization)

Lasso

$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s$$



Ridge and Lasso Regularization



A Simple Special Case for Ridge Regression and the Lasso

In order to obtain a better intuition about the behavior of ridge regression and the lasso, consider a simple special case with $n = p$, and $\mathbf{X}$ a diagonal matrix with 1's on the diagonal and 0's in all off-diagonal elements. To simplify the problem further, assume also that we are performing regression without an intercept. With these assumptions, the usual least squares problem simplifies to finding $\beta_1, \dots, \beta_p$ that minimize

$$\sum_{j=1}^p (y_j - \beta_j)^2. \quad (6.11)$$

In this case, the least squares solution is given by

$$\hat{\beta}_j = y_j.$$

A Simple Special Case for Ridge Regression and the Lasso

And in this setting, ridge regression amounts to finding $\beta_1, \dots, \beta_p$ such that

$$\sum_{j=1}^p (y_j - \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (6.12)$$

is minimized, and the lasso amounts to finding the coefficients such that

$$\sum_{j=1}^p (y_j - \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (6.13)$$

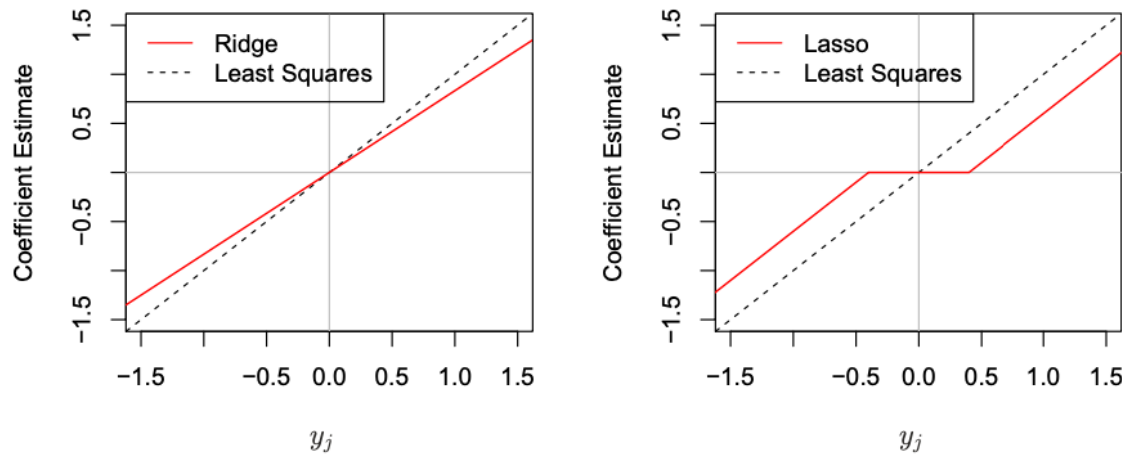
is minimized. One can show that in this setting, the ridge regression estimates take the form

$$\hat{\beta}_j^R = y_j / (1 + \lambda), \quad (6.14)$$

and the lasso estimates take the form

$$\hat{\beta}_j^L = \begin{cases} y_j - \lambda/2 & \text{if } y_j > \lambda/2; \\ y_j + \lambda/2 & \text{if } y_j < -\lambda/2; \\ 0 & \text{if } |y_j| \leq \lambda/2. \end{cases} \quad (6.15)$$

A Simple Special Case for Ridge Regression and the Lasso



is minimized. One can show that in this setting, the ridge regression estimates take the form

$$\hat{\beta}_j^R = y_j / (1 + \lambda), \quad (6.14)$$

and the lasso estimates take the form

$$\hat{\beta}_j^L = \begin{cases} y_j - \lambda/2 & \text{if } y_j > \lambda/2; \\ y_j + \lambda/2 & \text{if } y_j < -\lambda/2; \\ 0 & \text{if } |y_j| \leq \lambda/2. \end{cases} \quad (6.15)$$

l_q Penalties

For a fixed real number $q \geq 0$, consider the criterion

$$\underset{\beta \in \mathbb{R}^p}{\text{minimize}} \left\{ \frac{1}{2N} \sum_{i=1}^N (y_i - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|^q \right\}. \quad (2.21)$$

This is the lasso for $q = 1$ and ridge regression for $q = 2$. For $q = 0$, the term $\sum_{j=1}^p |\beta_j|^q$ counts the number of nonzero elements in β , and so solving (2.21) amounts to best-subset selection. Figure 2.6 displays the constraint regions corresponding to these penalties for the case of two predictors ($p = 2$). Both

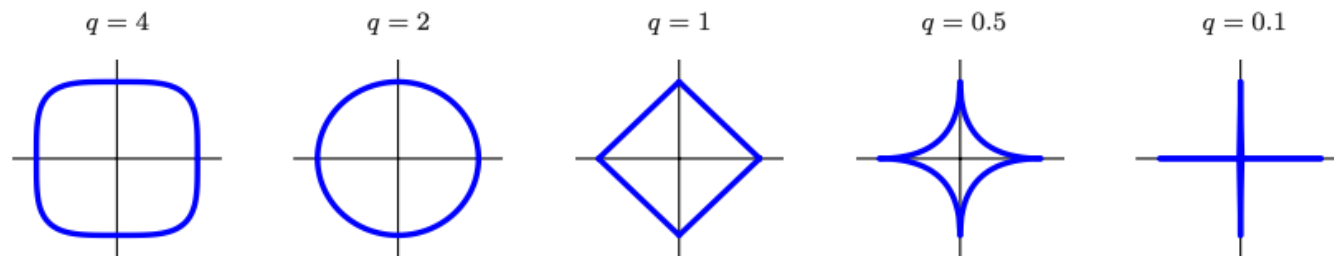


Figure 2.6 Constraint regions $\sum_{j=1}^p |\beta_j|^q \leq 1$ for different values of q . For $q < 1$, the constraint region is nonconvex.

Nonconvex Penalties

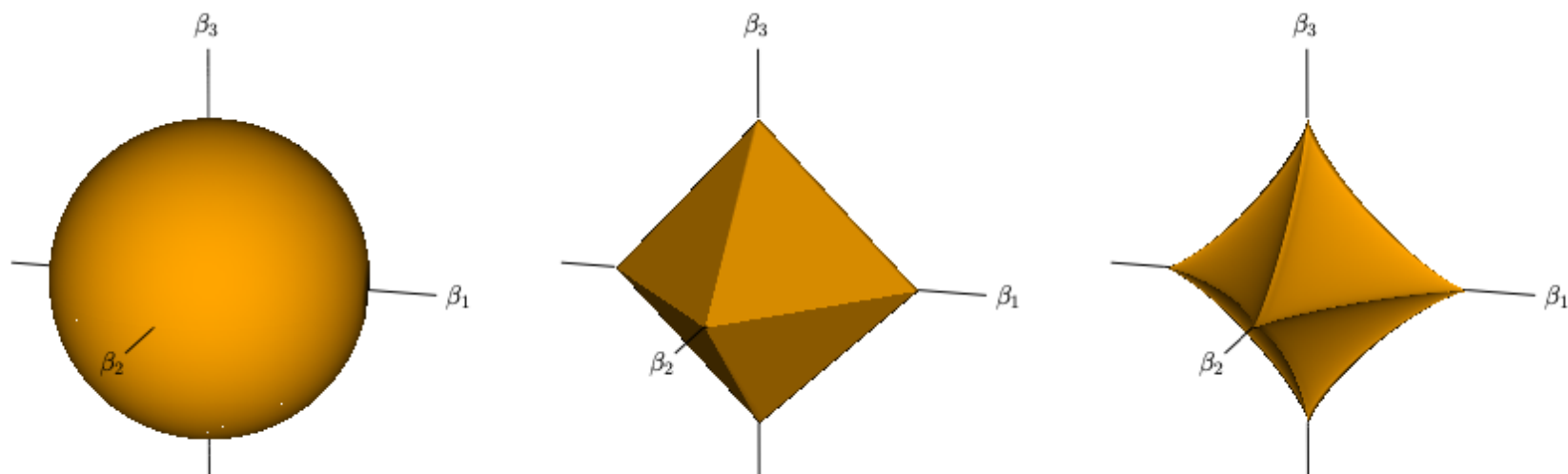


Figure 4.12 The ℓ_q unit balls in $\mathbb{R}^3$ for $q = 2$ (left), $q = 1$ (middle), and $q = 0.8$ (right). For $q < 1$ the constraint regions are nonconvex. Smaller q will correspond to fewer nonzero coefficients, and less shrinkage. The nonconvexity leads to combinatorially hard optimization problems.

Elastic net

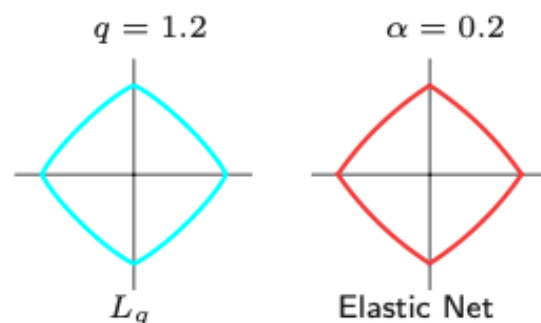
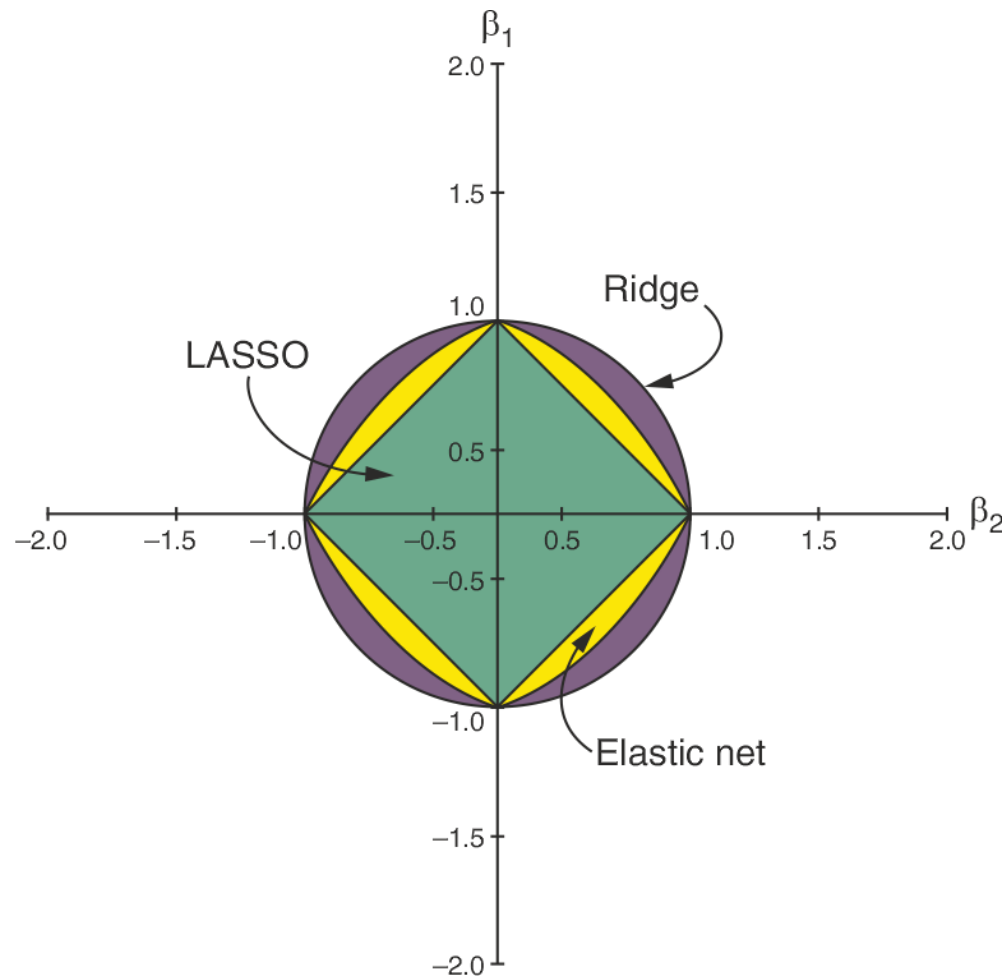


FIGURE 3.13. Contours of constant value of $\sum_j |\beta_j|^q$ for $q = 1.2$ (left plot), and the elastic-net penalty $\sum_j (\alpha \beta_j^2 + (1 - \alpha) |\beta_j|)$ for $\alpha = 0.2$ (right plot). Although visually very similar, the elastic-net has sharp (non-differentiable) corners, while the $q = 1.2$ penalty does not.

setting coefficients exactly to zero. Partly for this reason as well as for computational tractability, Zou and Hastie (2005) introduced the *elastic-net* penalty

$$\lambda \sum_{j=1}^p (\alpha \beta_j^2 + (1 - \alpha) |\beta_j|), \quad (3.54)$$

Ridge, Lasso and Elastic net Regularization



Elastic net and Lasso Regularization

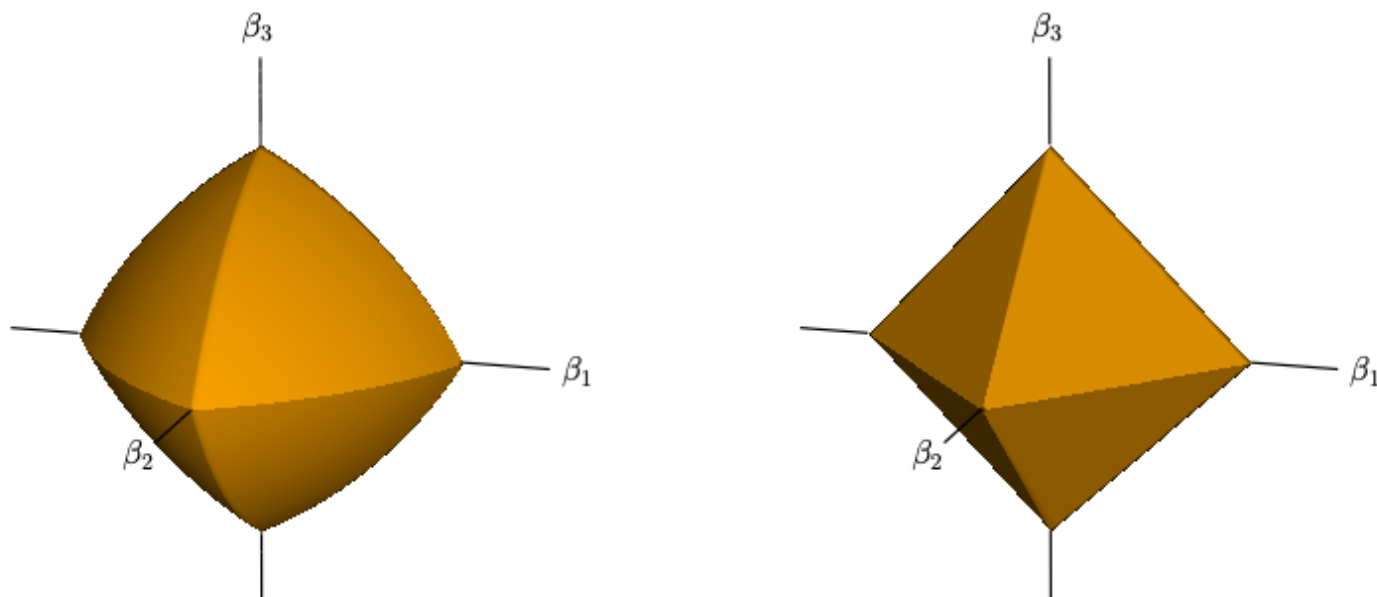


Figure 4.2 The elastic-net ball with $\alpha = 0.7$ (left panel) in $\mathbb{R}^3$, compared to the ℓ_1 ball (right panel). The curved contours encourage strongly correlated variables to share coefficients (see Exercise 4.2 for details).